Acidity, Basicity, and p*K*a

- **Acid-Base equilibria**

Using Organometallic reagents to make C-C bonds

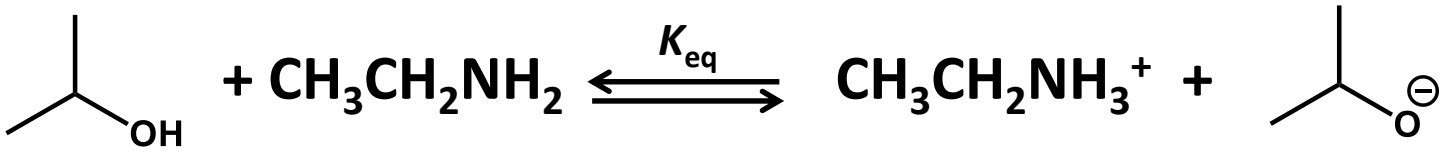
- **Organometallics are nucleophilic and often strongly basic**
- **Making organometallics from halo-compounds**
- **Making organometallics by deprotonating carbon atoms**
- **Using organometallics to make new C-C bonds from C=O groups**

Acidity, Basicity, and p*K*a

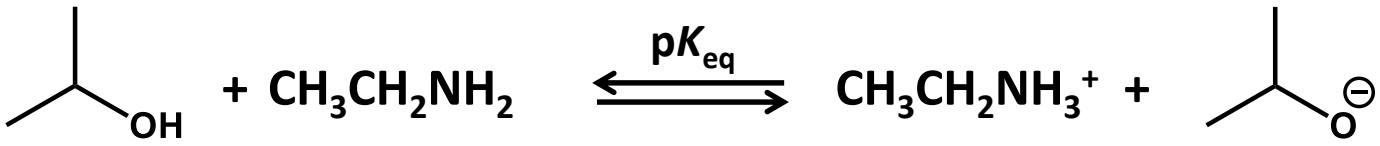
- **Acid-Base equilibria**

Acid-Base equilibria

- **Since pK_a always refers to the ionization of an acid in water, it can be used to predict the position of acid-base equilibria**

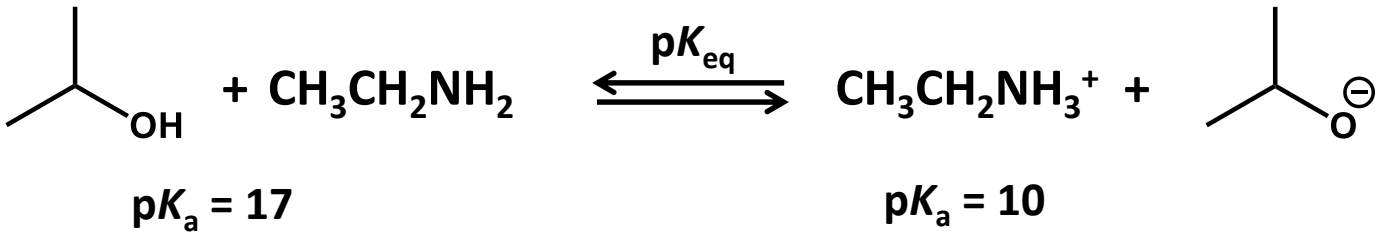


Acid-Base equilibria



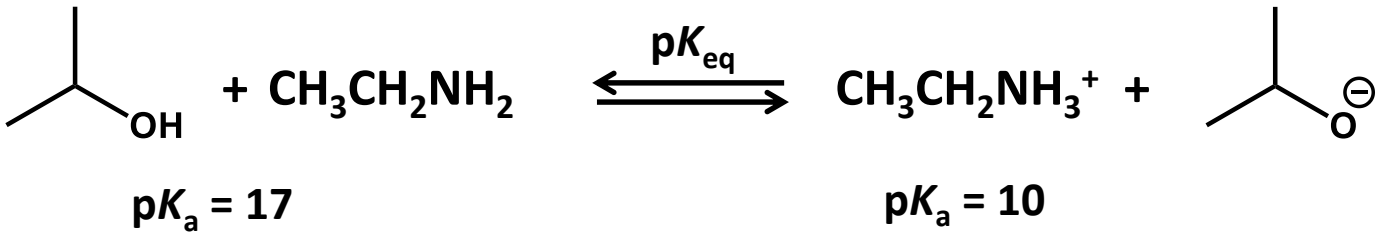
$pK_a = 17$ $pK_a = 10$

Acid-Base equilibria



- **To evaluate the position of the equilibrium, pK_{eq} = pK_a (reactant acid) – pK_a (product acid)**

Acid-Base equilibria



- **To evaluate the position of the equilibrium, pK_{eq} = pK_a (reactant acid) – pK_a (product acid)**

Here 17 -10 = 7, so $K_{eq} = 10^{-7}$

No reaction (no ammonium)

Acid-Base equilibria

- **What about the reaction of MeLi and water?**

Acid-Base equilibria

- **What about the reaction of MeLi and water?**
  - **pK_a H_2O = 15.74**
  - **pK_a CH_4 = 50**

Acid-Base equilibria

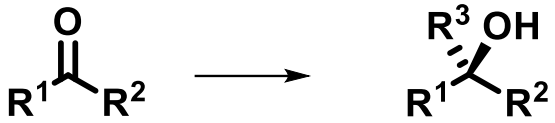
- **What about the reaction of MeLi and water?**
  - **pK_a H_2O = 15.74**
  - **pK_a CH_4 = 50**



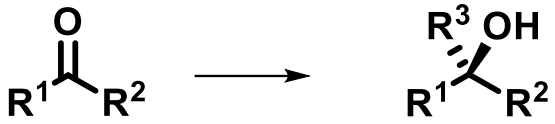
$$K_{eq} = 10^{34.3}$$

Quantitative reaction

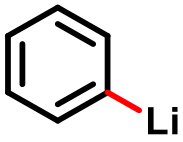
C-C bond formation using organometallic reagents-Revision (Lecture 6)



C-C bond formation using organometallic reagents-Revision (Lecture 6)

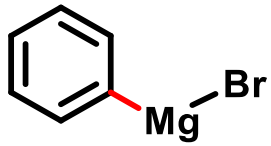


Examples of nucleophilic sigma bonds: Organometallic Reagents



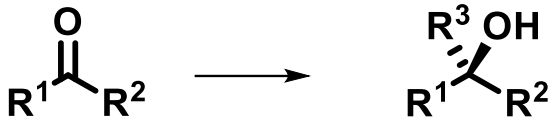




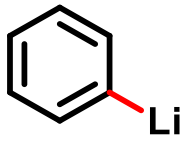




C-C bond formation using organometallic reagents-Revision (Lecture 6)

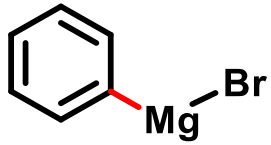


Examples of nucleophilic sigma bonds: Organometallic Reagents

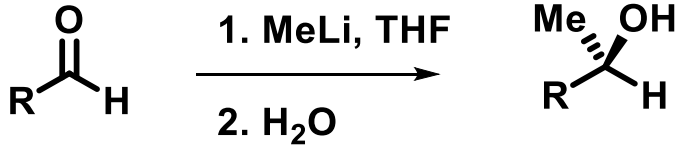






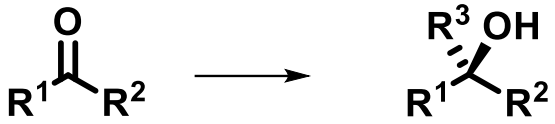




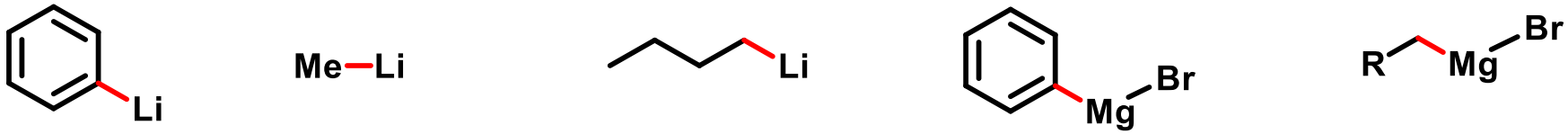


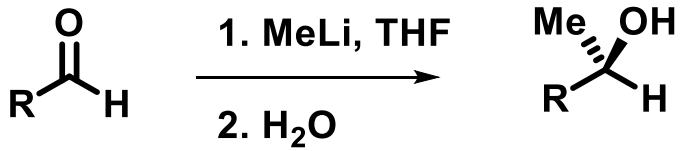
Reaction

C-C bond formation using organometallic reagents-Revision (Lecture 6)

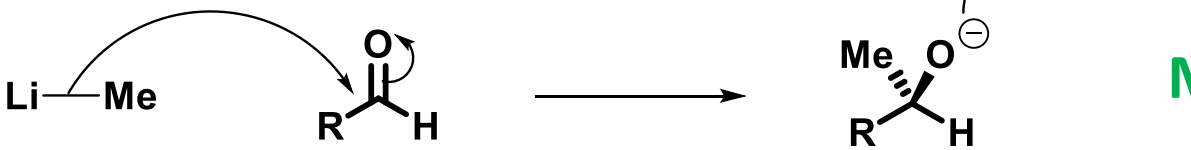


Examples of nucleophilic sigma bonds: Organometallic Reagents





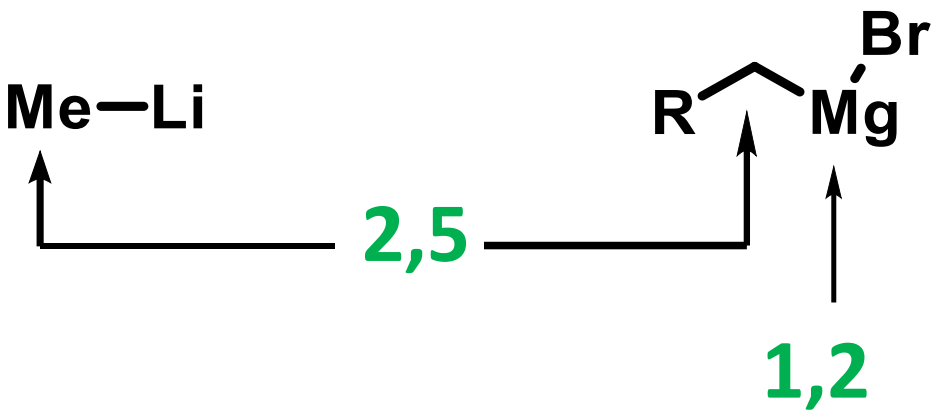
Reaction



Mechanism

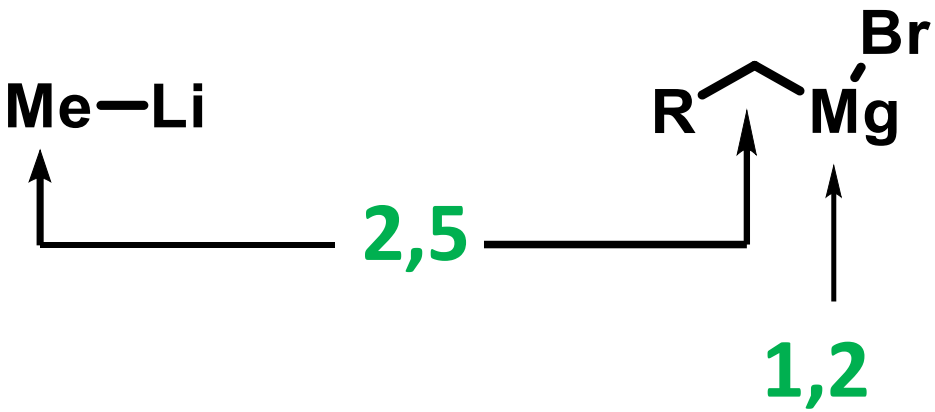
C-C bond formation using organometallic reagents

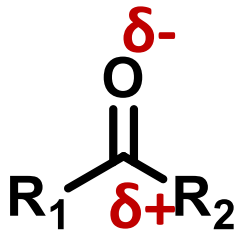
Electronegativities

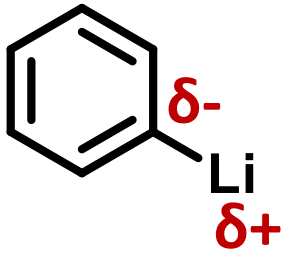


C-C bond formation using organometallic reagents

Electronegativities

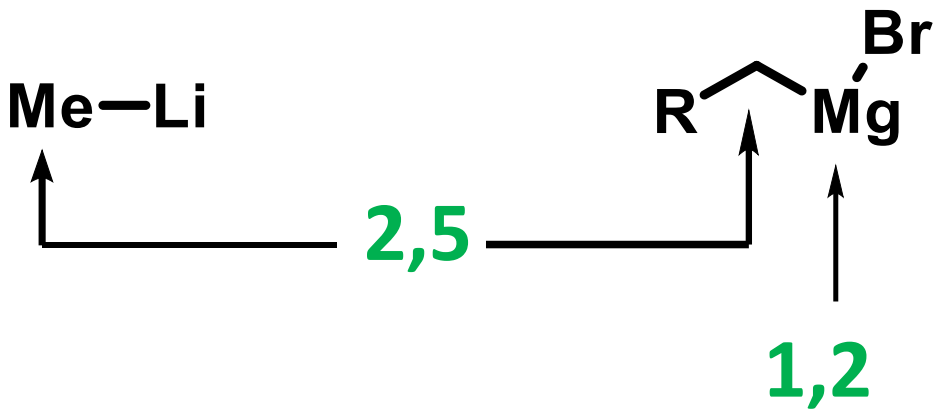


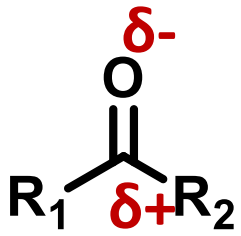


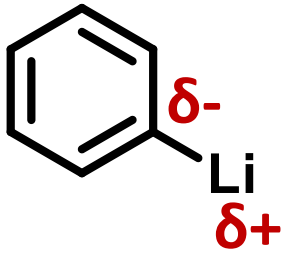


C-C bond formation using organometallic reagents

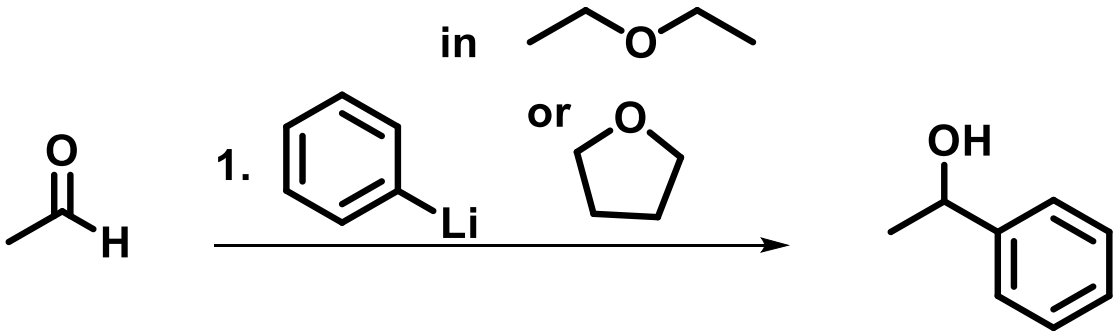
Electronegativities





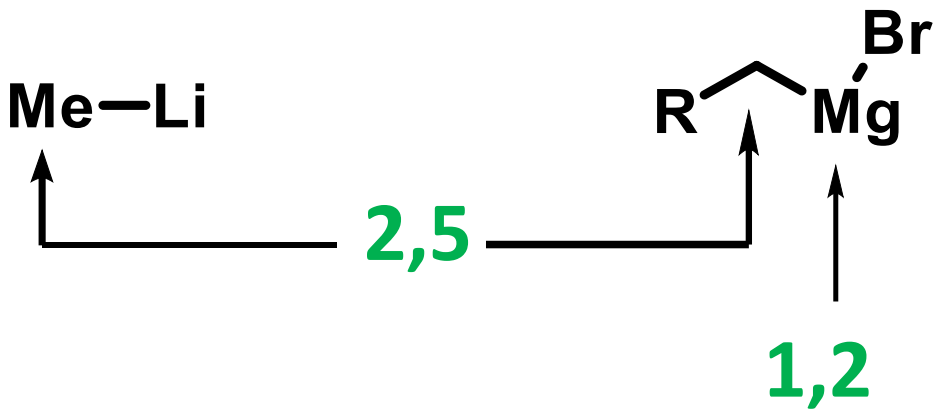


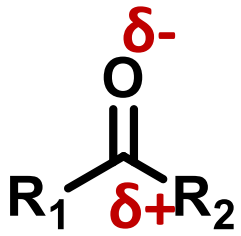
Example

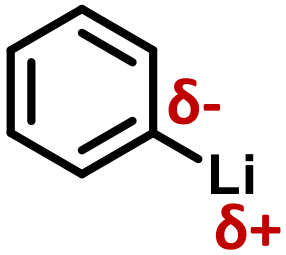


C-C bond formation using organometallic reagents

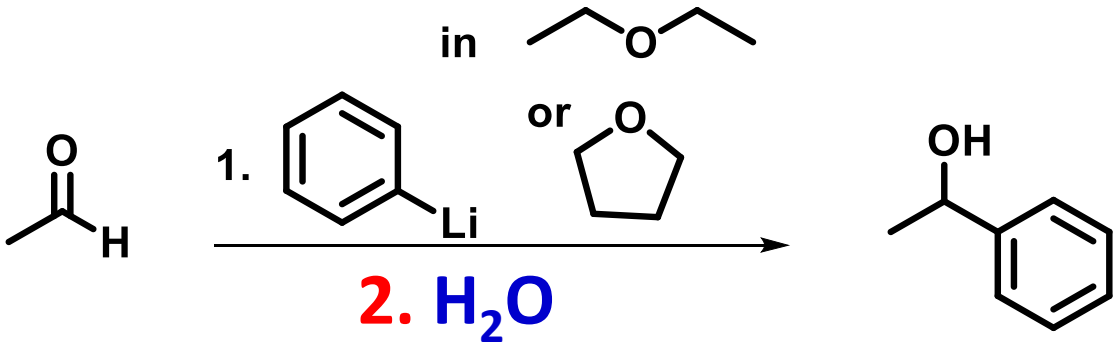
Electronegativities





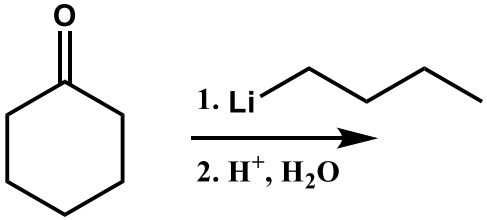


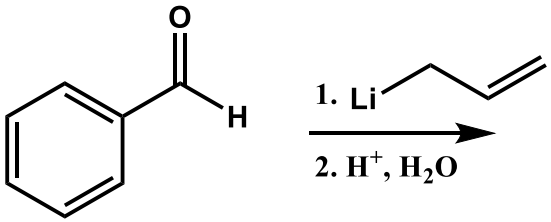
Example

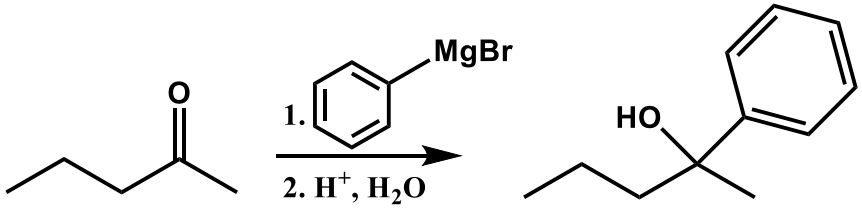


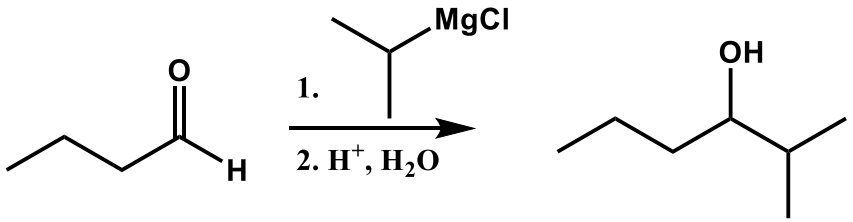
Water MUST NOT be present during the 1st step of the reaction

Draw mechanisms for these reactions!

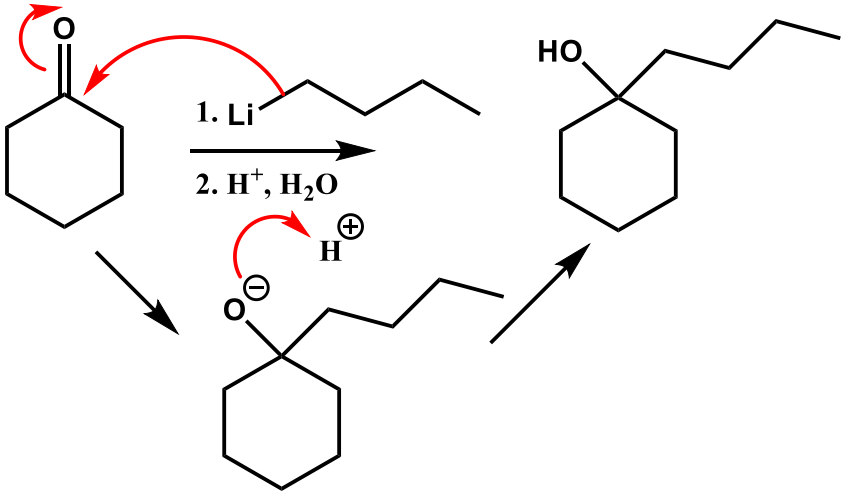


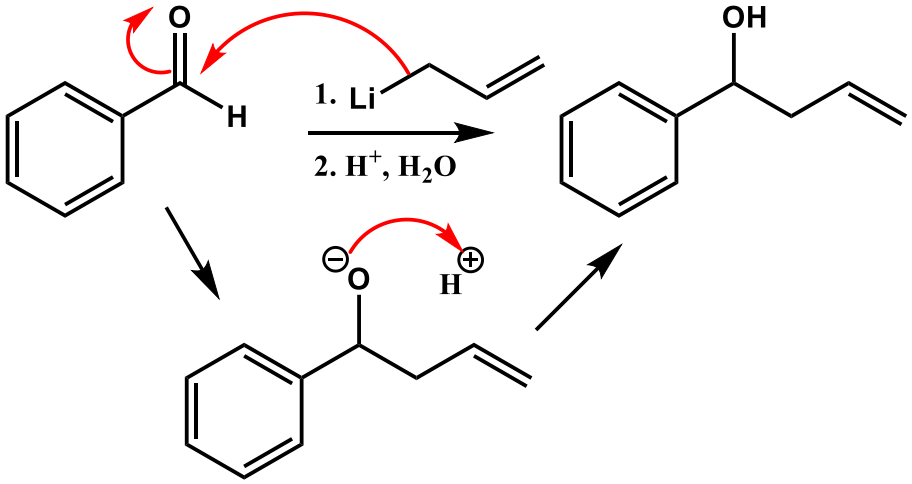


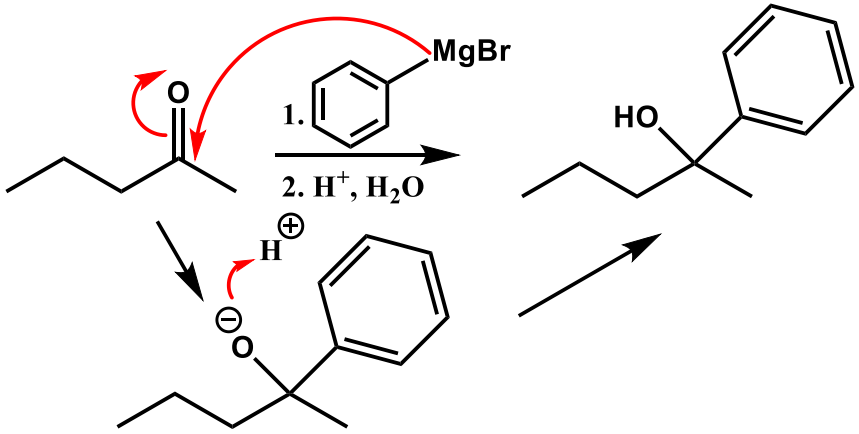


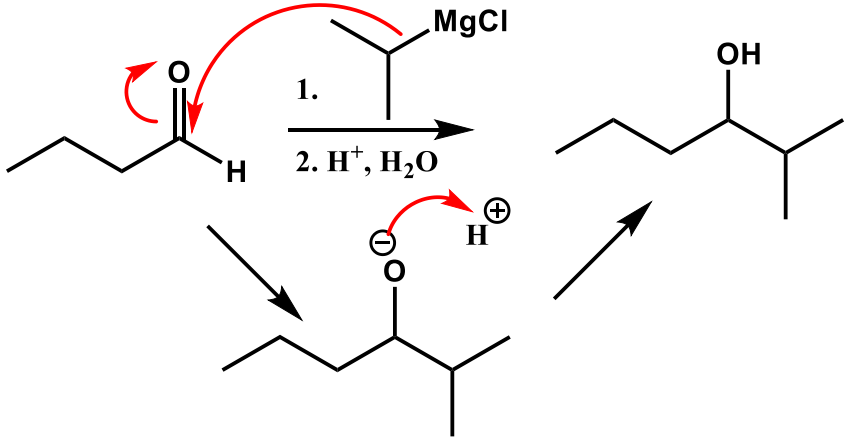


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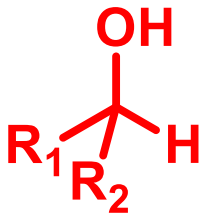


Outline

- **What kind of molecules we can make of organometallics**
- **What kind of molecules react with organometallics**
- **How to make organometallics**

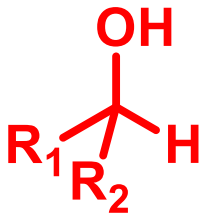
What kind of organic molecules can we make by using organometallics?

- **Secondary alcohols:**

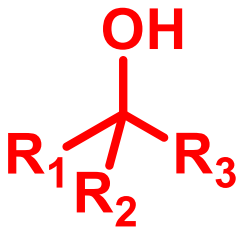


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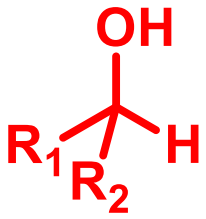


- **Tertiary alcohols:**

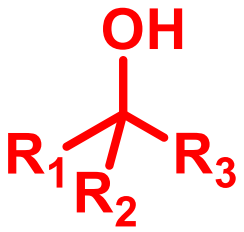


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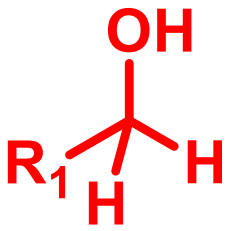
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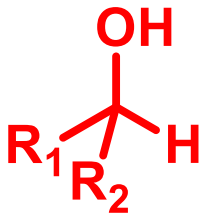


- **Primary alcohols:**

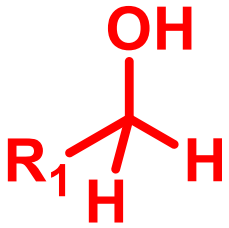


What kind of organic molecules can we make by using organometallics?

- **Secondary alcohols:**



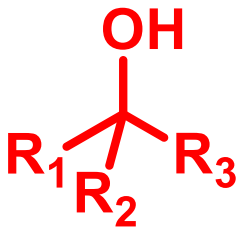
- **Primary alcohols:**



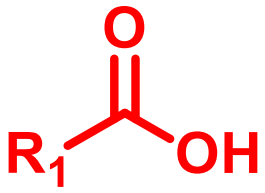
- **Alkanes:**



- **Tertiary alcohols:**



- **Carboxylic acids:**

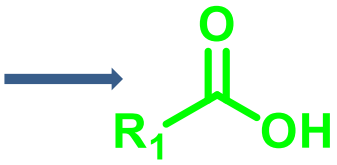


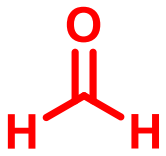
and many more…

What kind of molecules react with organometallics?

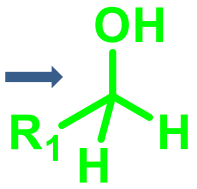
O=C=O

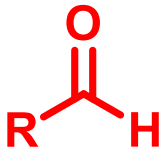
- Addition to CO_2 gives carboxylic acids



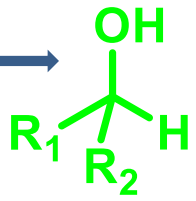


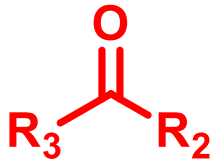
- Addition to formaldehyde (CH_2O) gives primary alcohols



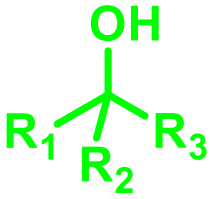


- Addition to other aldehydes gives secondary alcohols





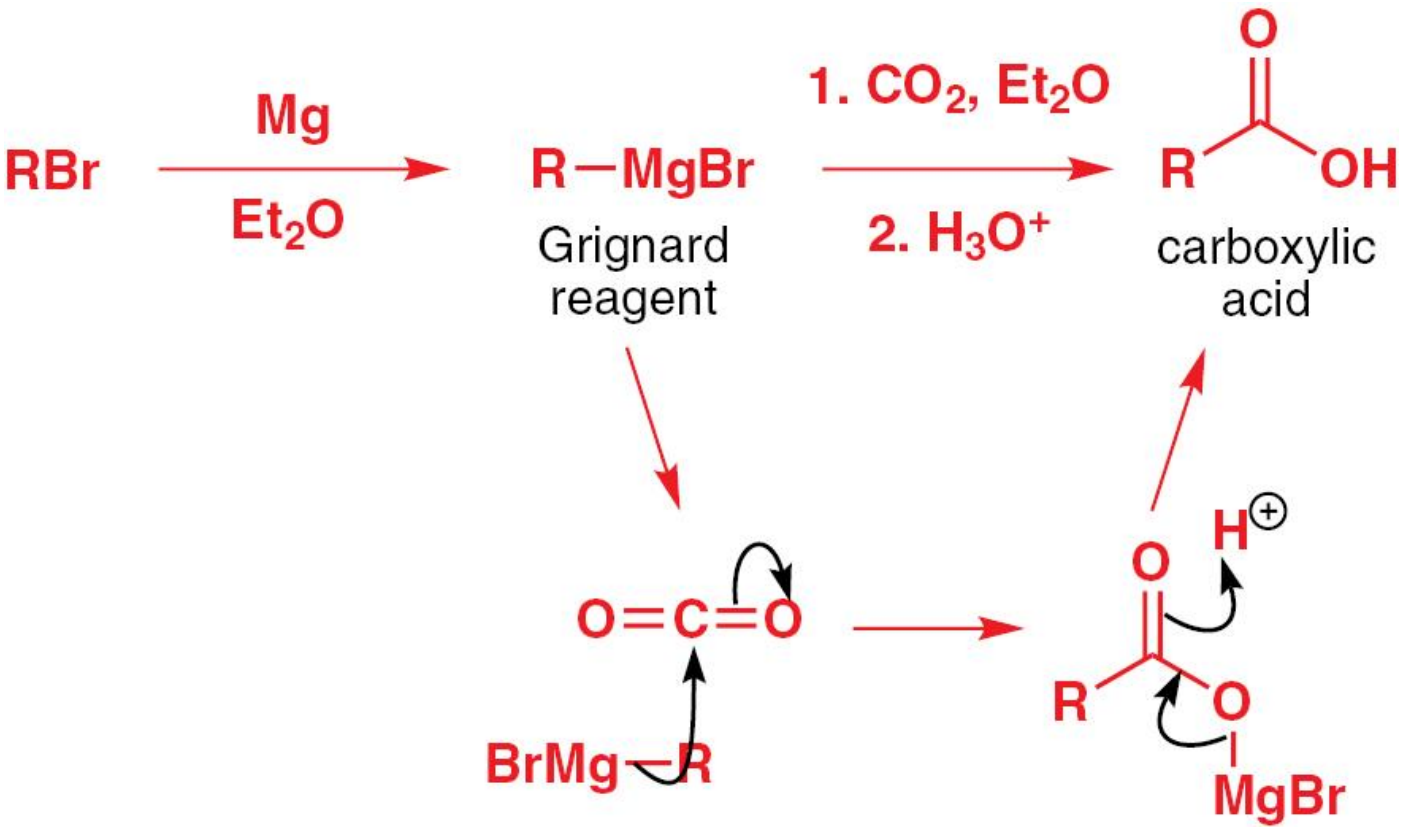
- Addition to ketones gives tertiary alcohols



Carboxylic acids from organometallics



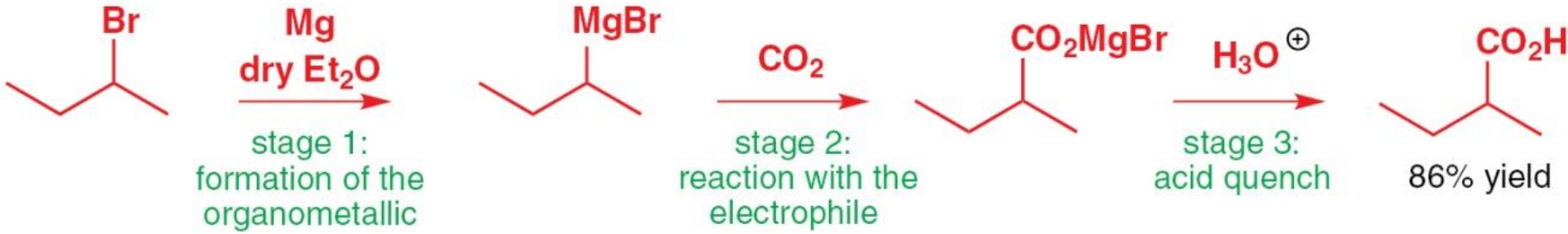
Carboxylic acids from organometallics



Carboxylic acids from organometallics

Examples

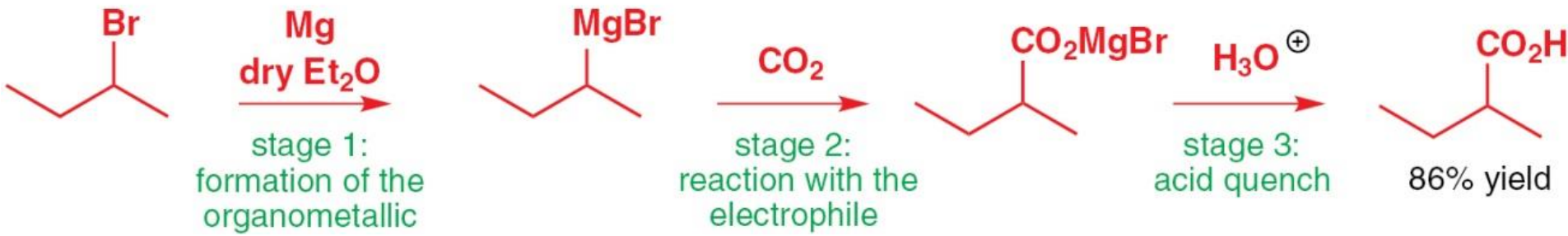
carboxylic acids from organometallics

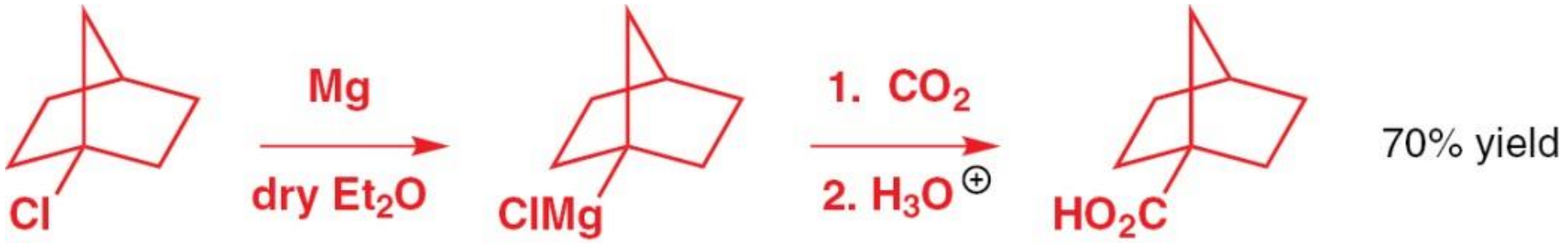


Carboxylic acids from organometallics

Examples

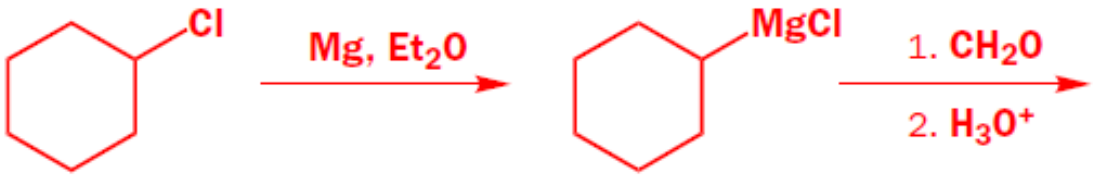
carboxylic acids from organometallics





Primary alcohols from organometallics

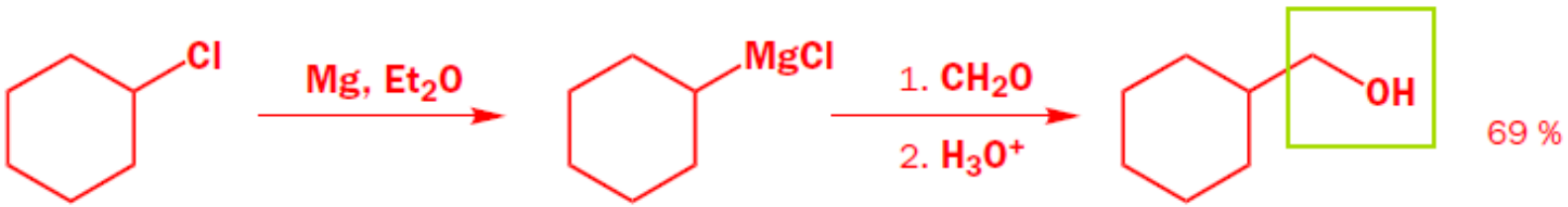
a primary alcohol from formaldehyde



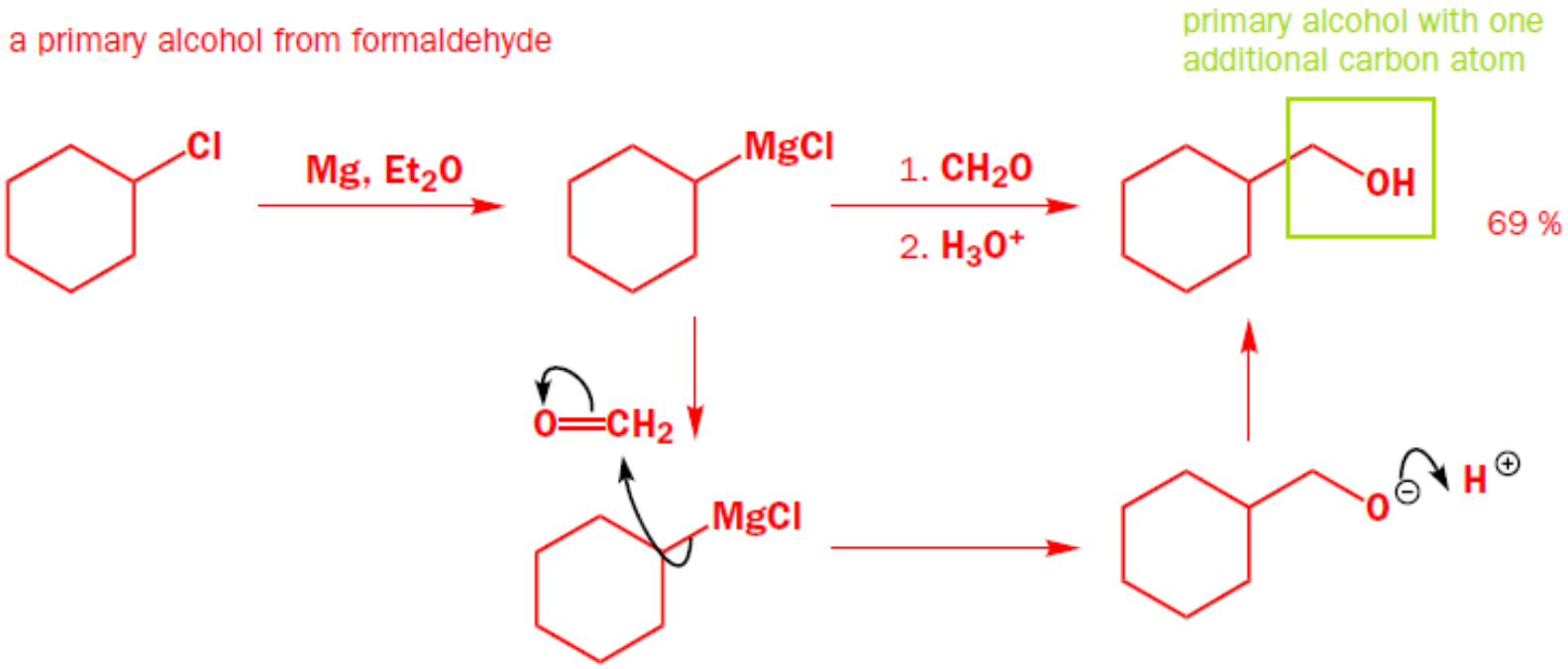
Primary alcohols from organometallics

a primary alcohol from formaldehyde

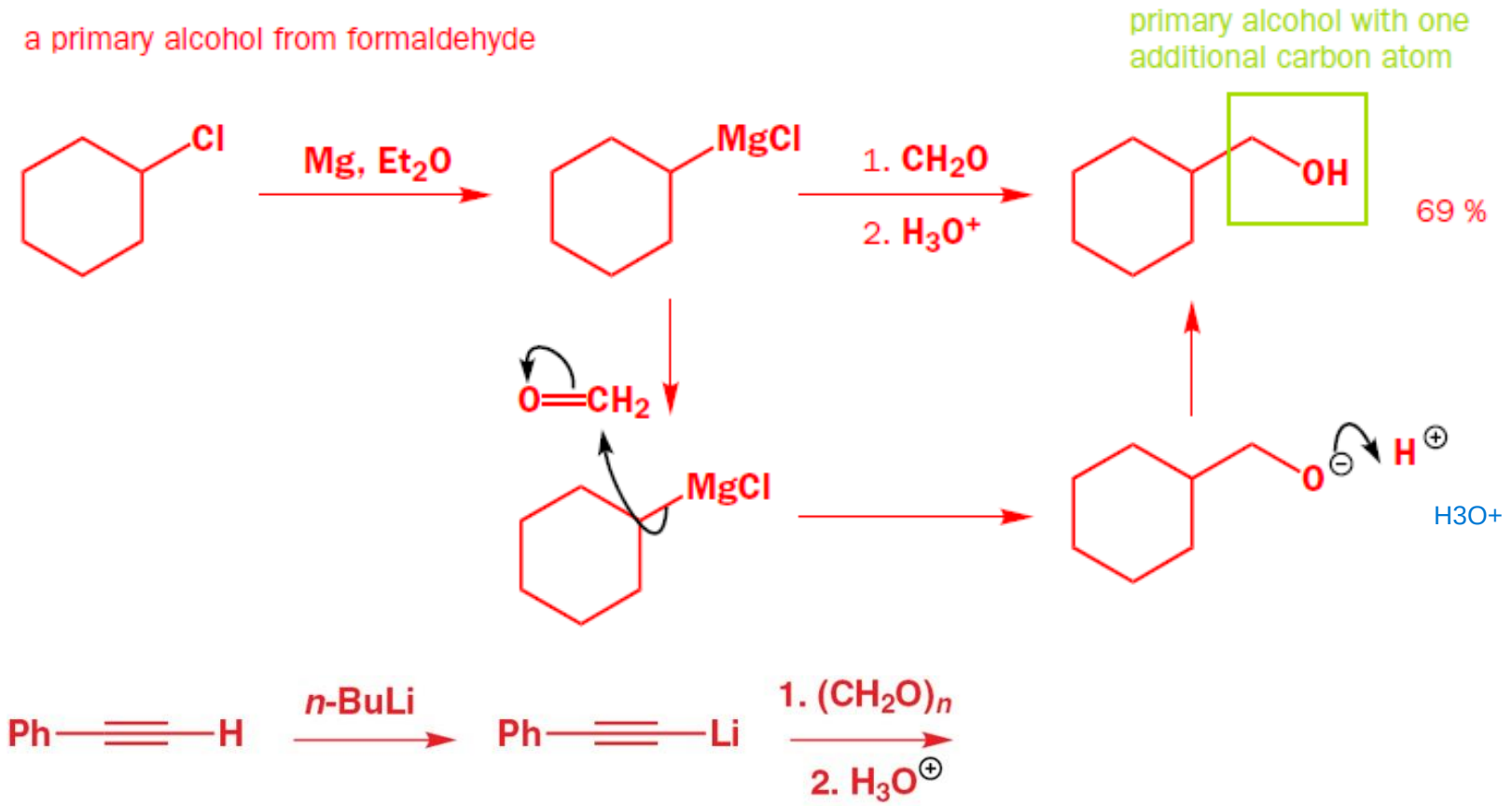
primary alcohol with one additional carbon atom



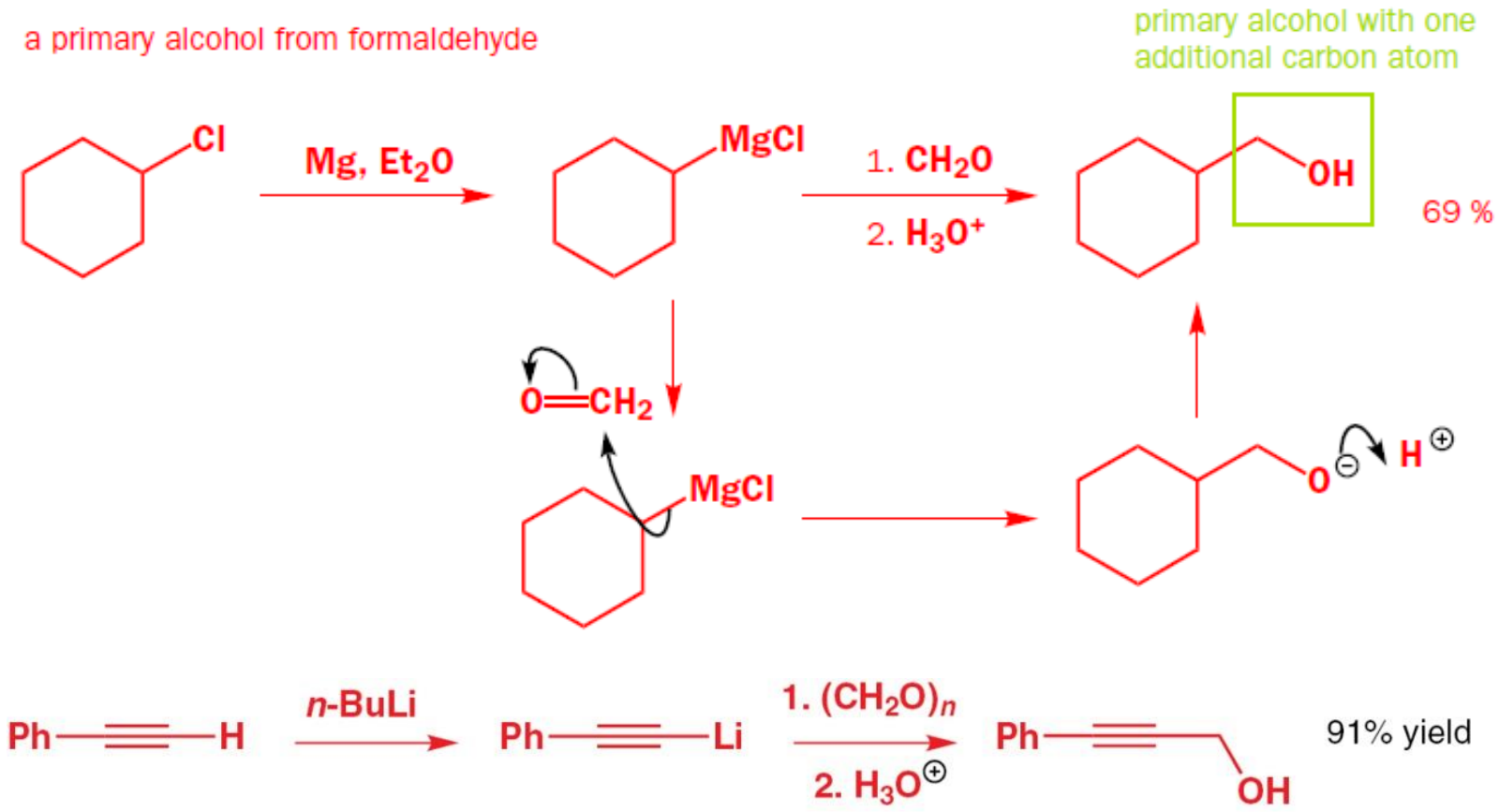
Primary alcohols from organometallics



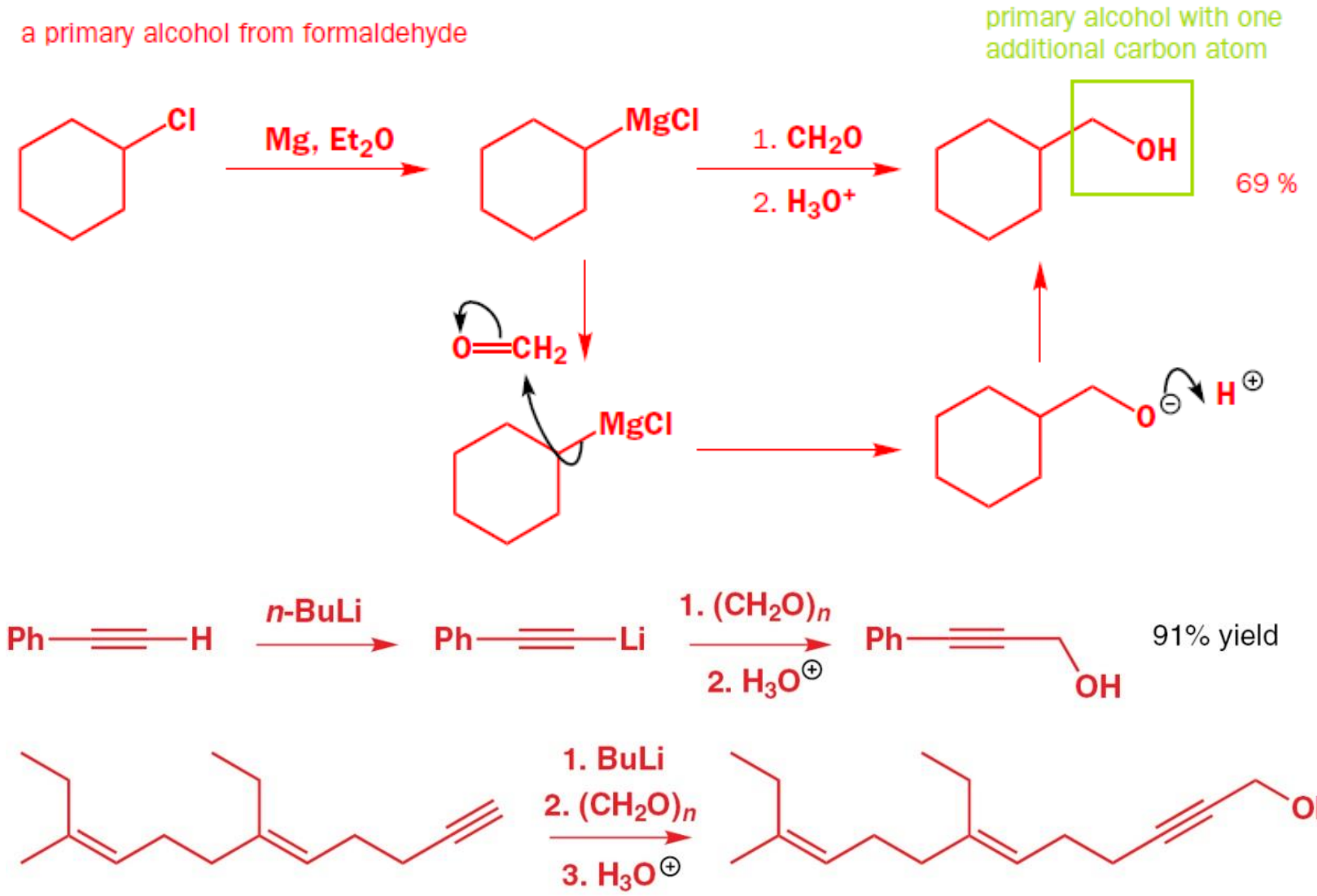
Primary alcohols from organometallics



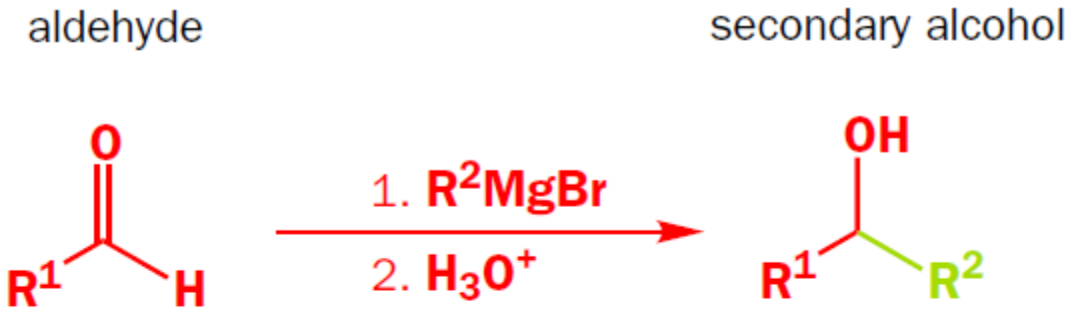
Primary alcohols from organometallics



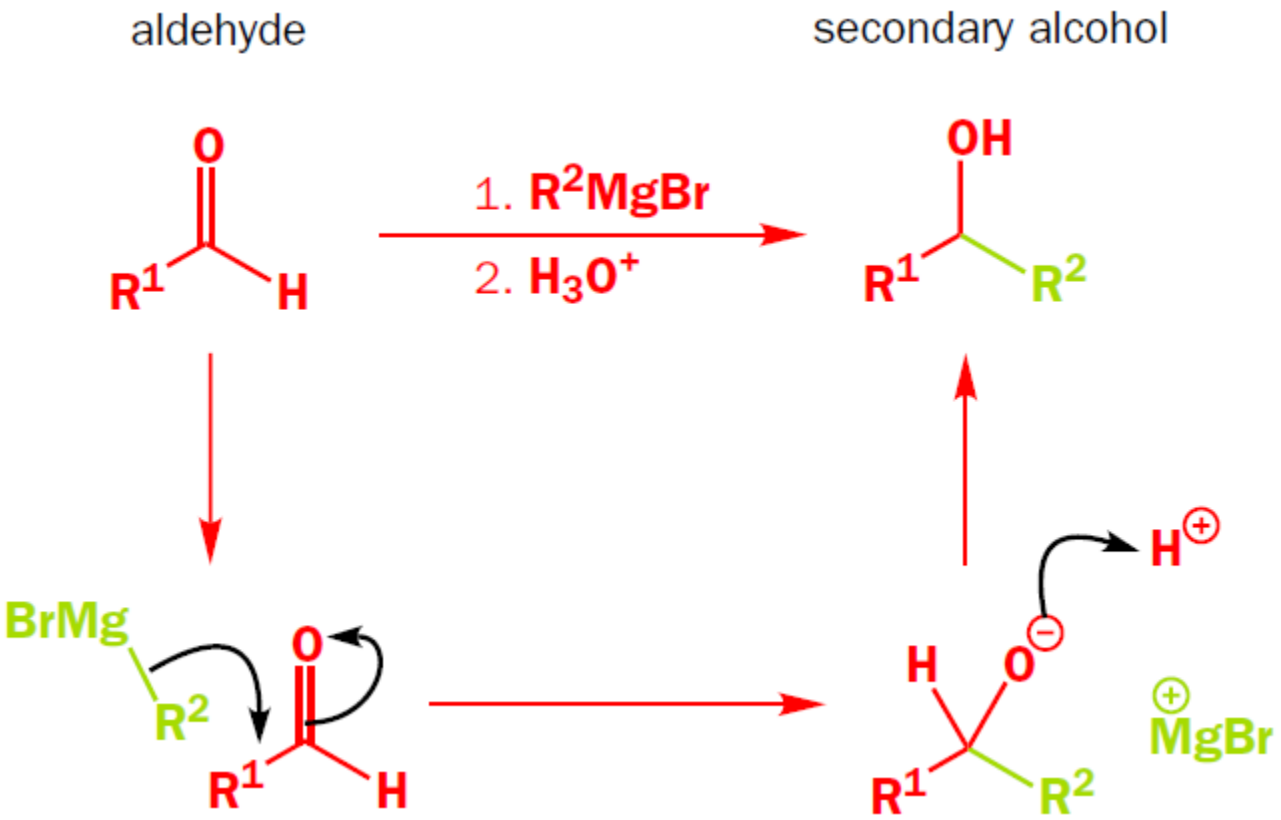
Primary alcohols from organometallics



Secondary alcohols from organometallics

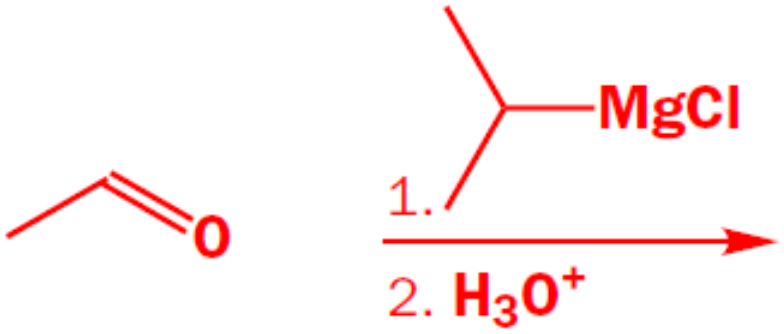


Secondary alcohols from organometallics



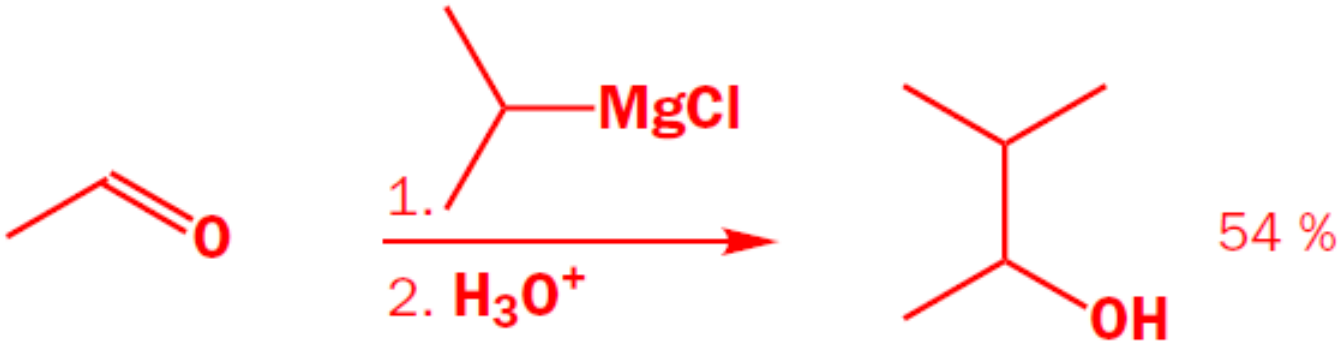
Secondary alcohols from organometallics

Examples



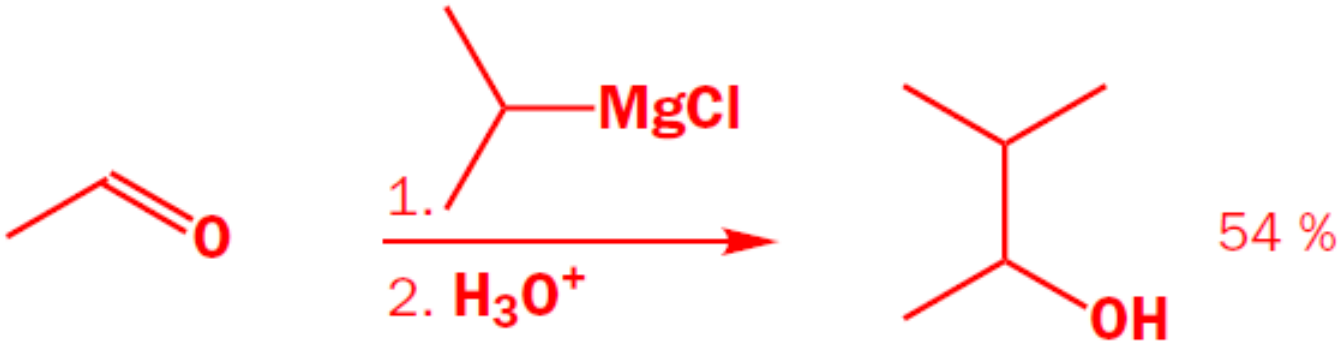
Secondary alcohols from organometallics

Examples



Secondary alcohols from organometallics

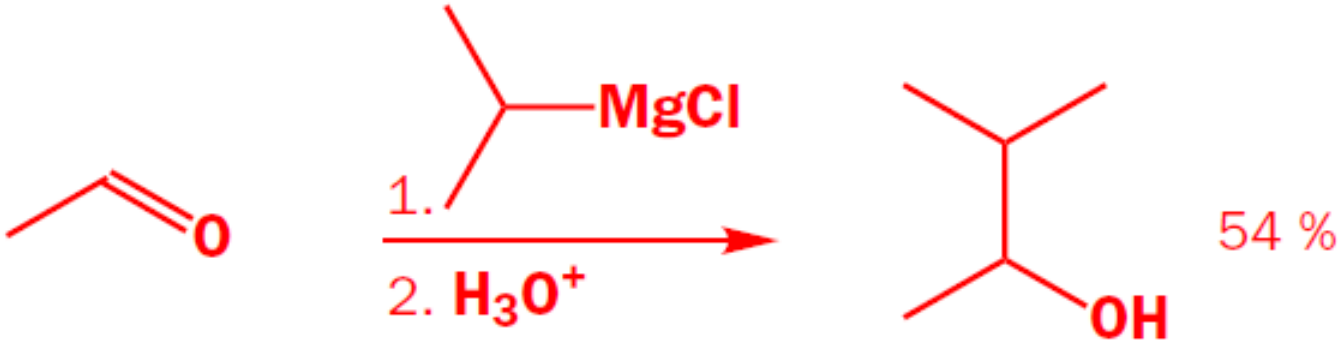
Examples

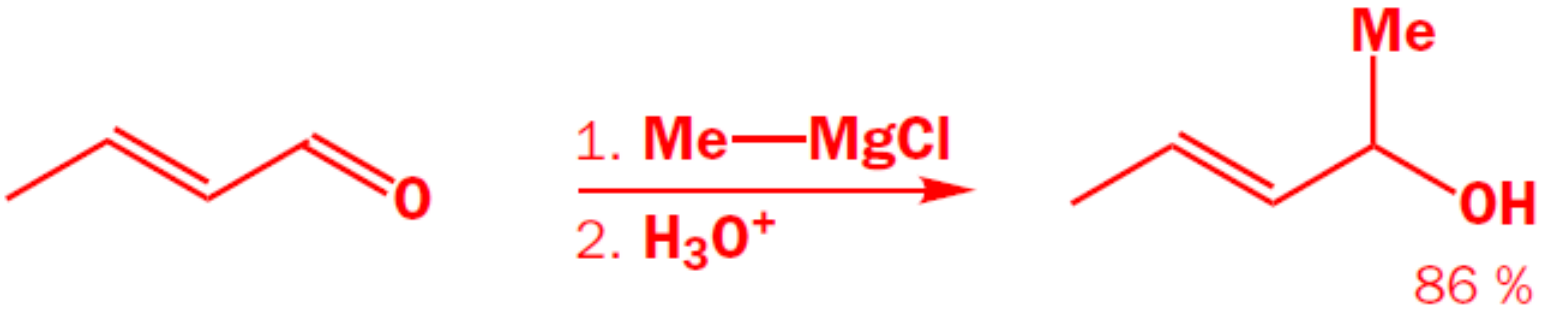




Secondary alcohols from organometallics

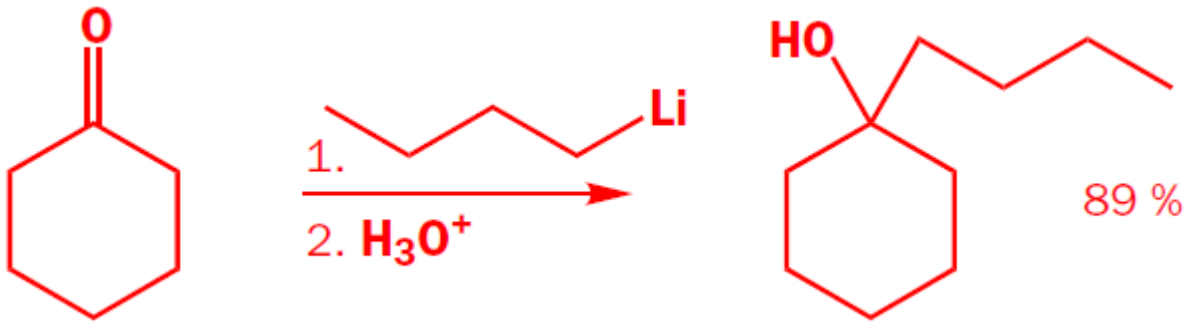
Examples





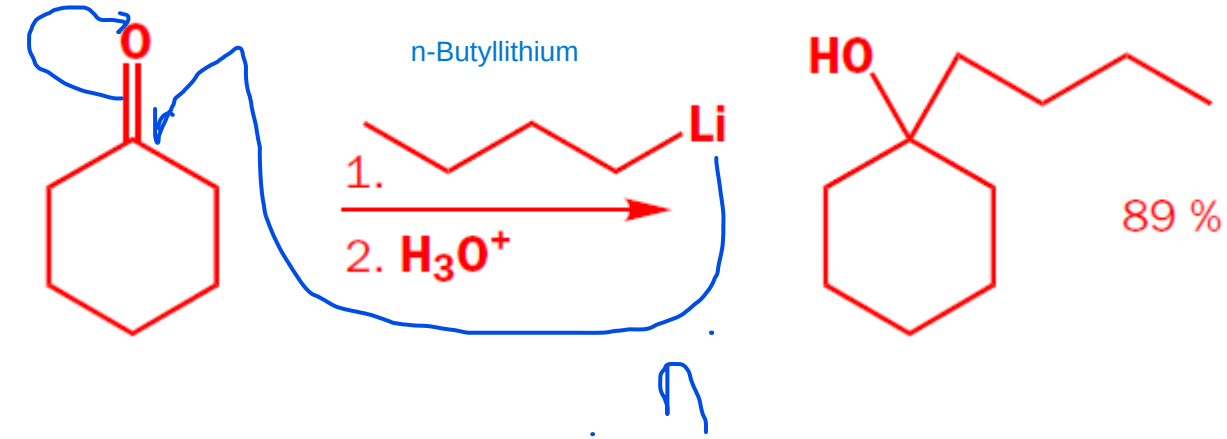
Tertiary alcohols from organometallics

Examples



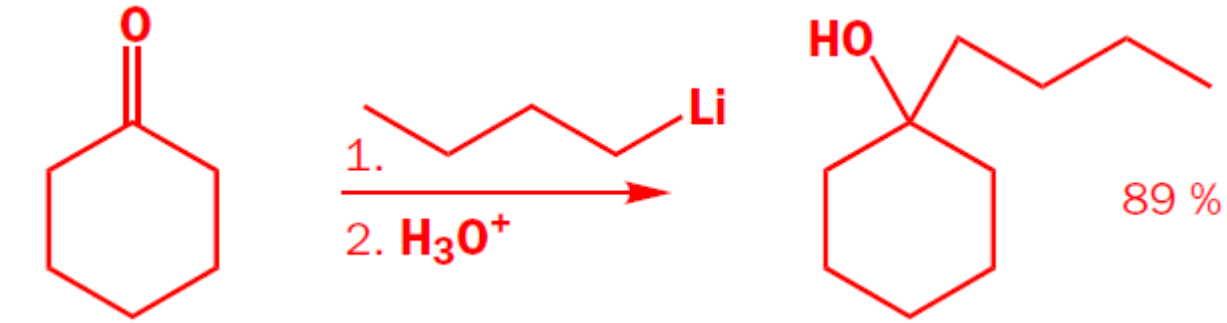
Tertiary alcohols from organometallics

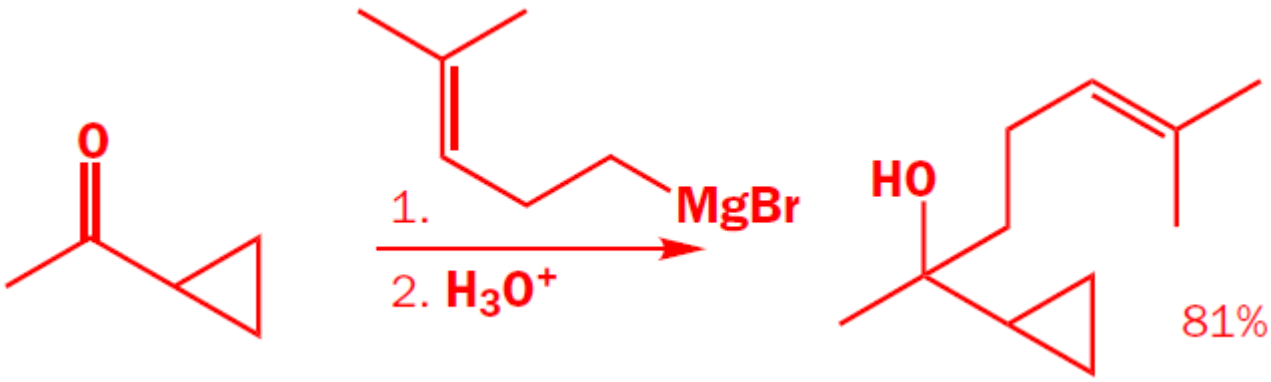
Examples



Tertiary alcohols from organometallics

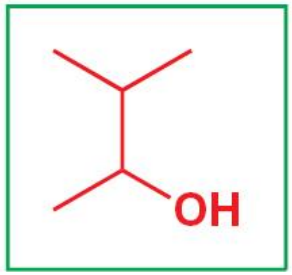
Examples

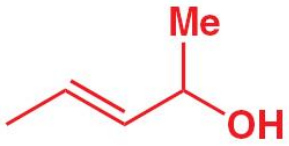




Which aldehyde? Which ketone?

Examples

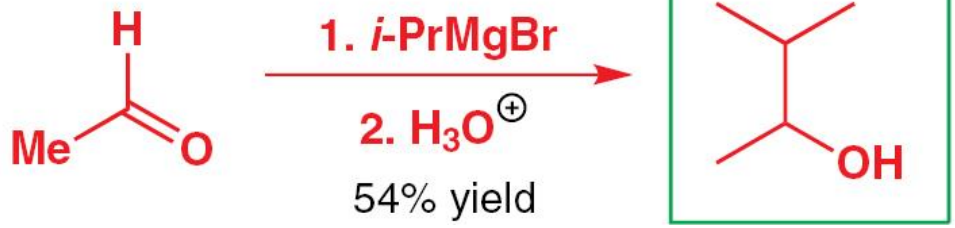


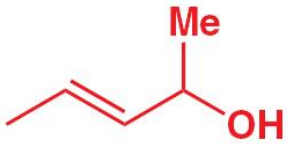


Which aldehyde? Which ketone?

Examples

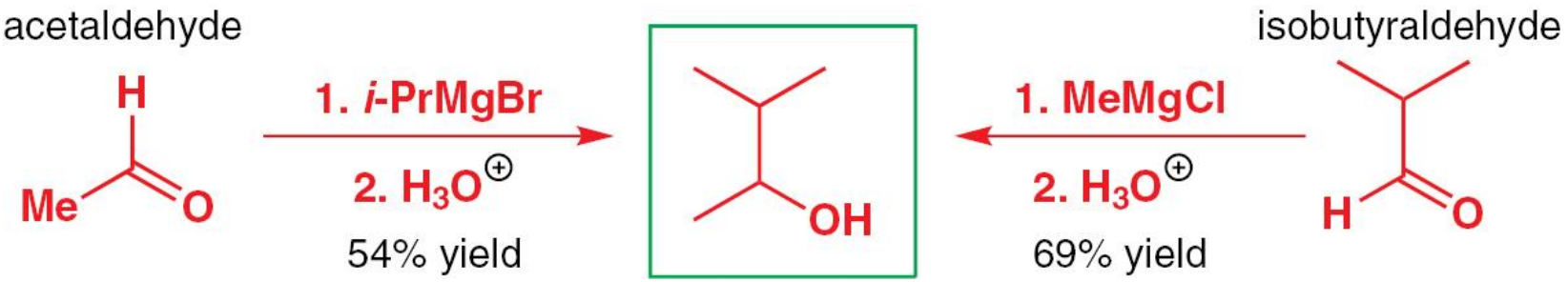
acetaldehyde

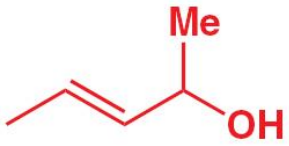




Which aldehyde? Which ketone?

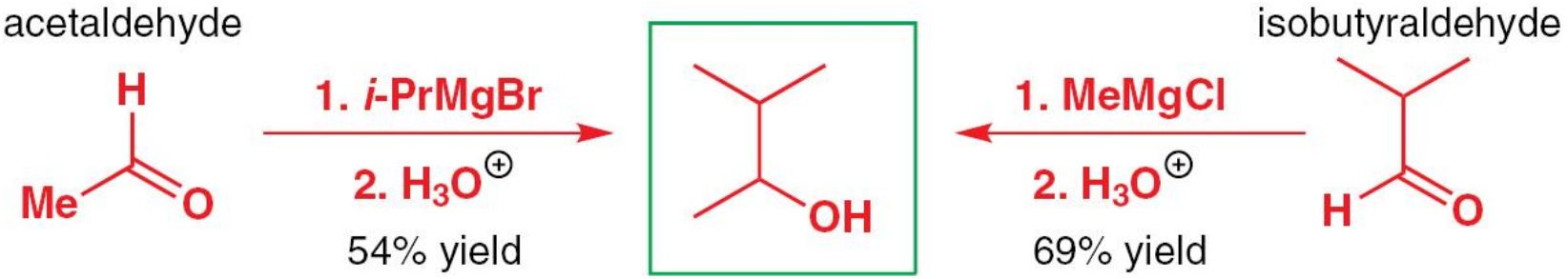
Examples

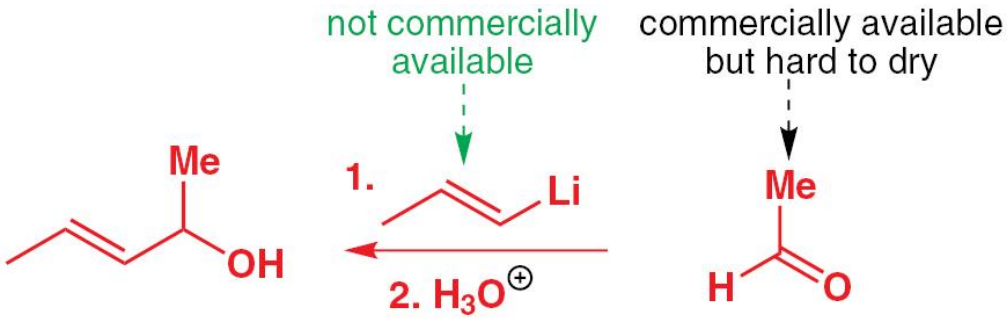




Which aldehyde? Which ketone?

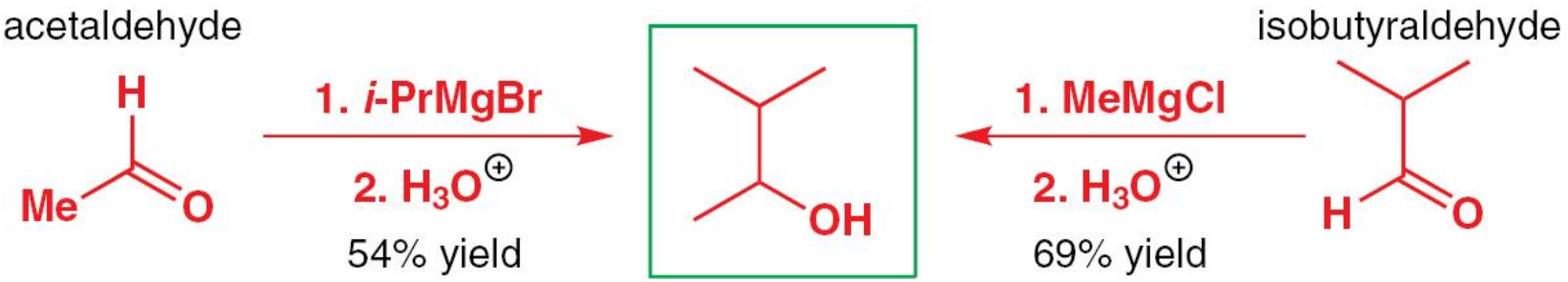
Examples

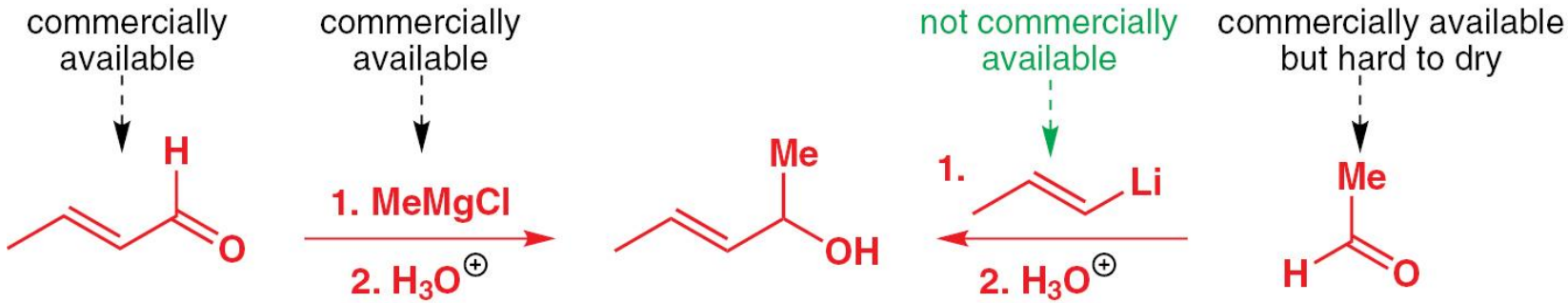




Which aldehyde? Which ketone?

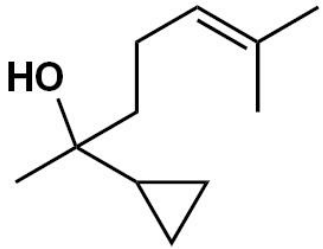
Examples





Which aldehyde? Which ketone?

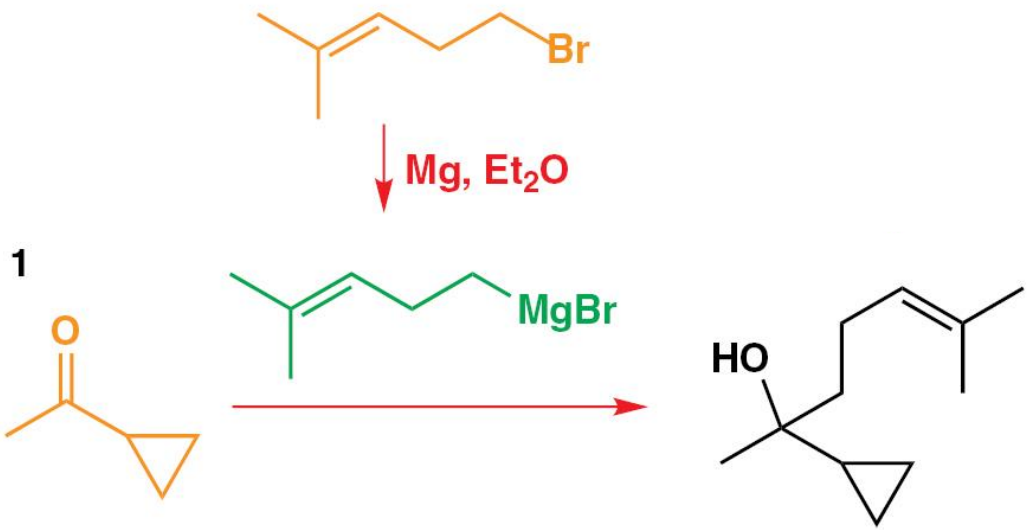
Examples



Commercially available reagents

Which aldehyde? Which ketone?

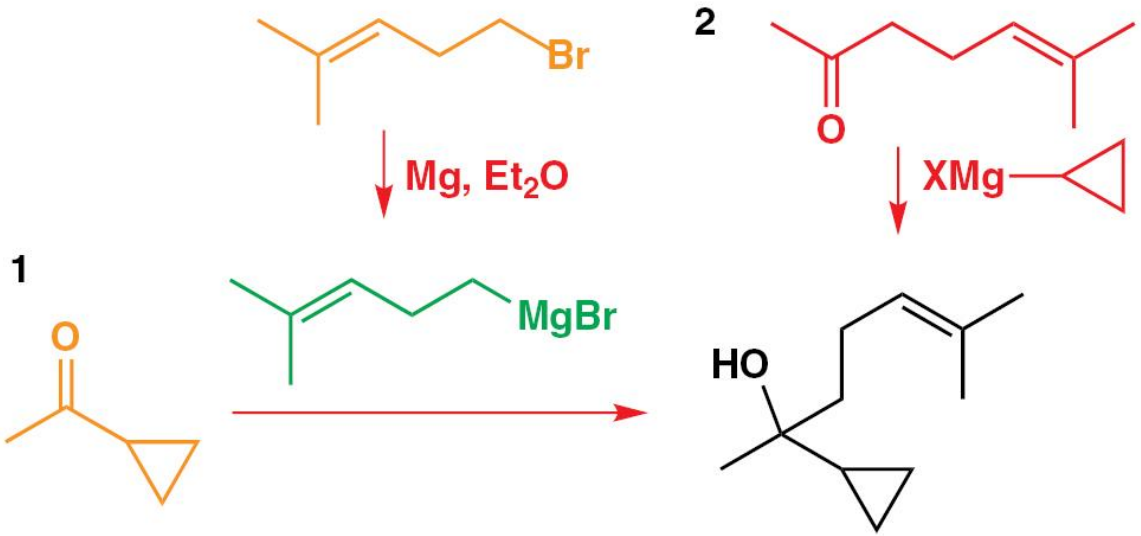
Examples



Commercially available reagents

Which aldehyde? Which ketone?

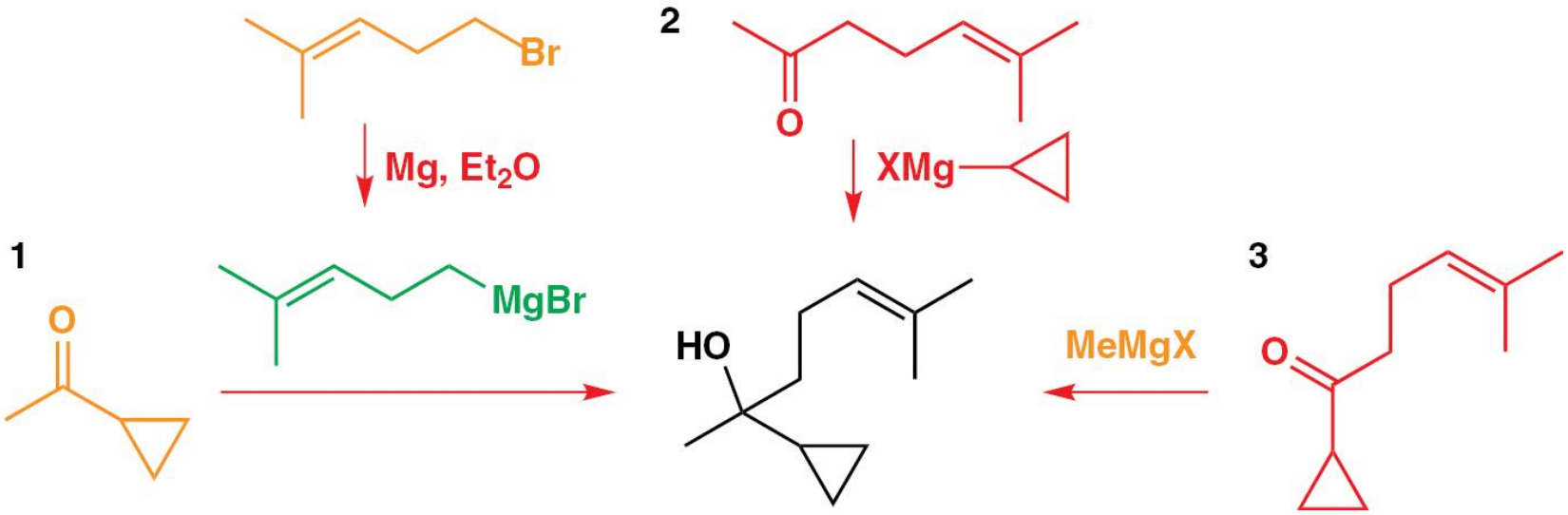
Examples



Commercially available reagents

Which aldehyde? Which ketone?

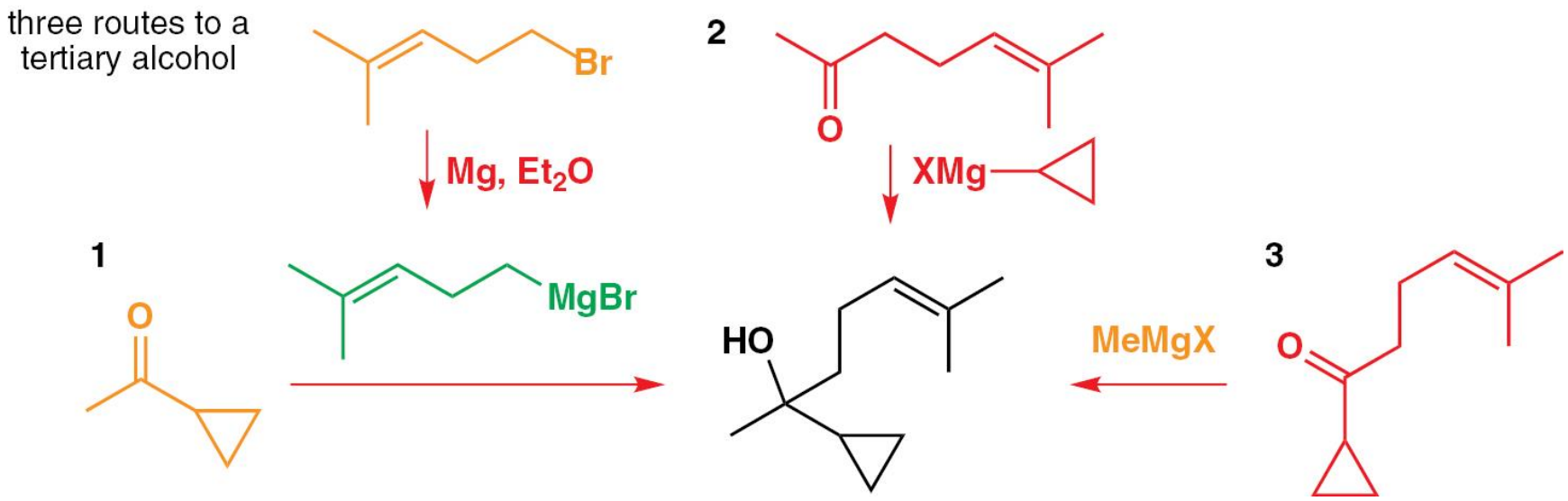
Examples



Commercially available reagents

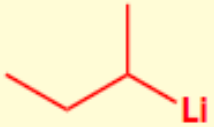
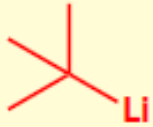
Which aldehyde? Which ketone?

Examples

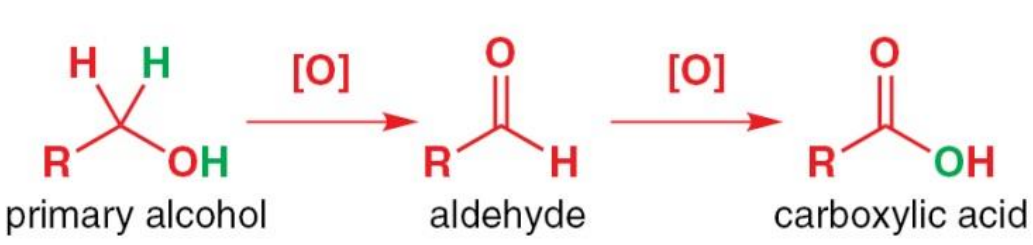


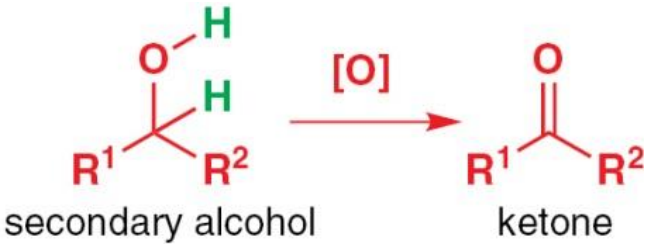
Commercially available reagents

Commercially available organometallics

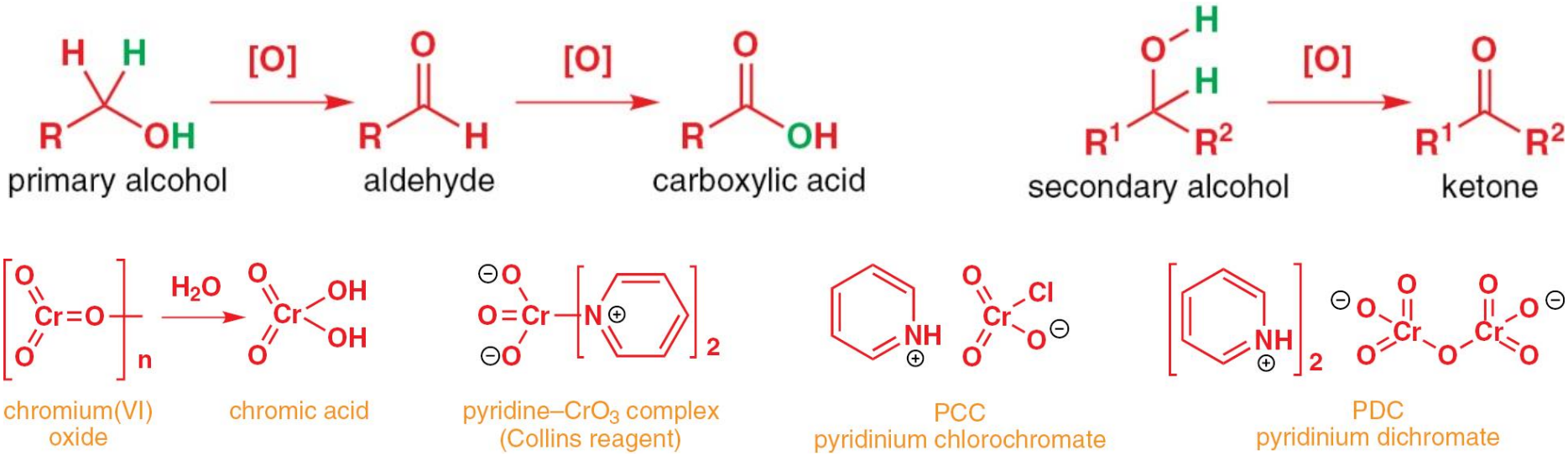
methyllithium (MeLi)	methylmagnesium chloride, bromide, and iodide (MeMgX)
n-butyllithium (*n*-BuLi or just BuLi) 	ethylmagnesium bromide (EtMgBr)
sec-butyllithium (*sec*-BuLi or *s*-BuLi) 	butylmagnesium chloride (BuMgCl)
tert-butyllithium (*tert*-BuLi or *t*-BuLi) 	allylmagnesium chloride and bromide 
phenyllithium (PhLi)	phenylmagnesium chloride and bromide (PhMgCl or PhMgBr)

Oxidation of alcohols: Back to aldehydes and ketones

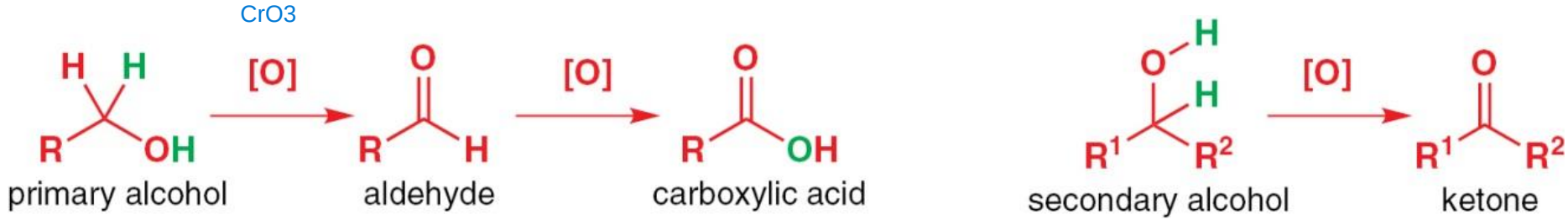


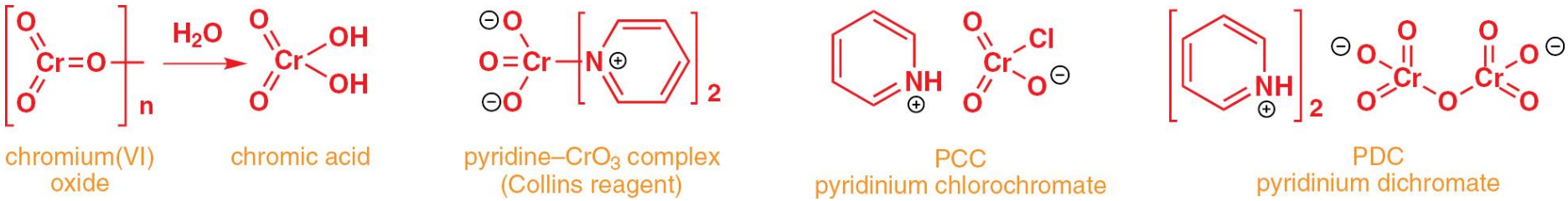


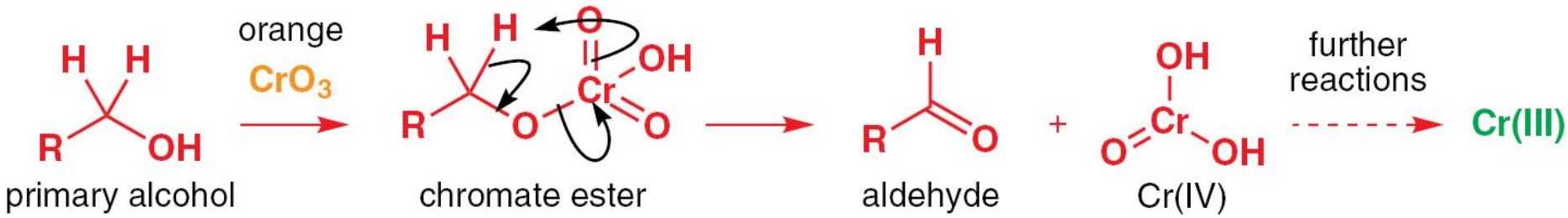
Oxidation of alcohols: Back to aldehydes and ketones



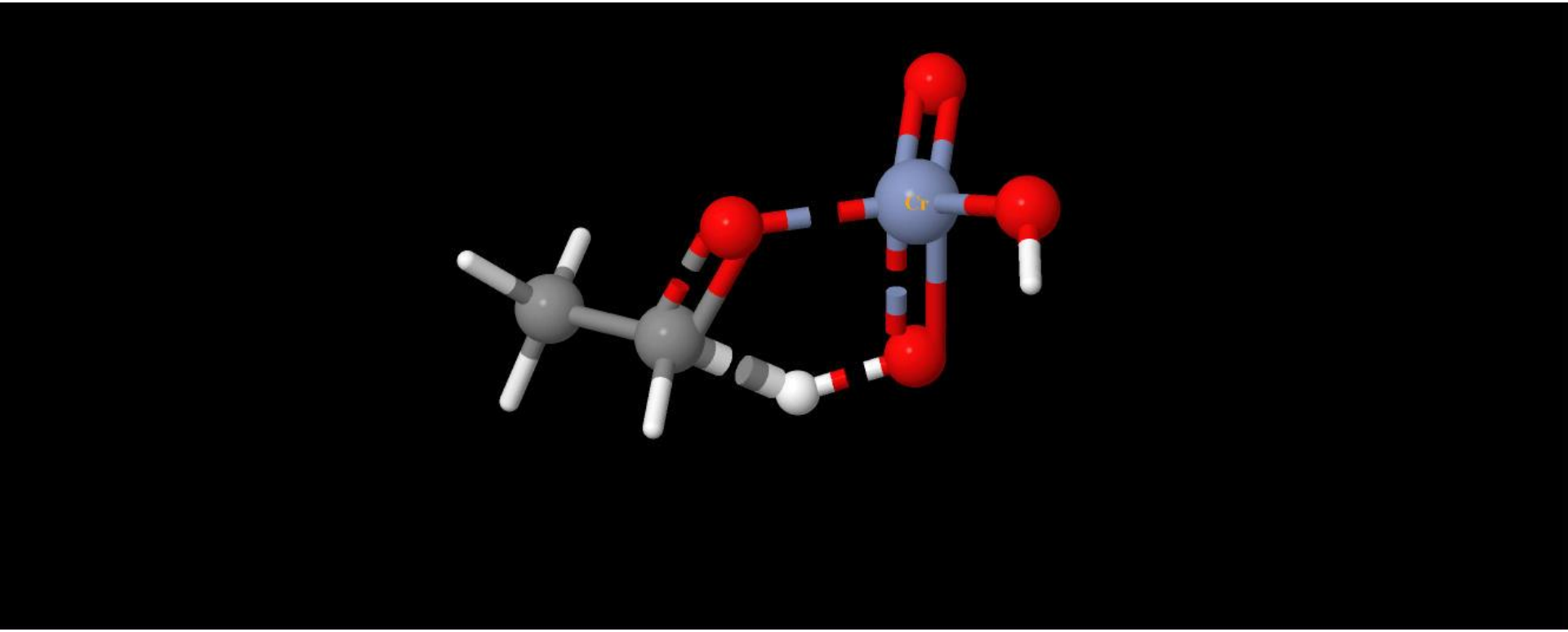
Oxidation of alcohols: Back to aldehydes and ketones







Ethanol oxidation



Visit: www.chemtube3d.com/clayden/195

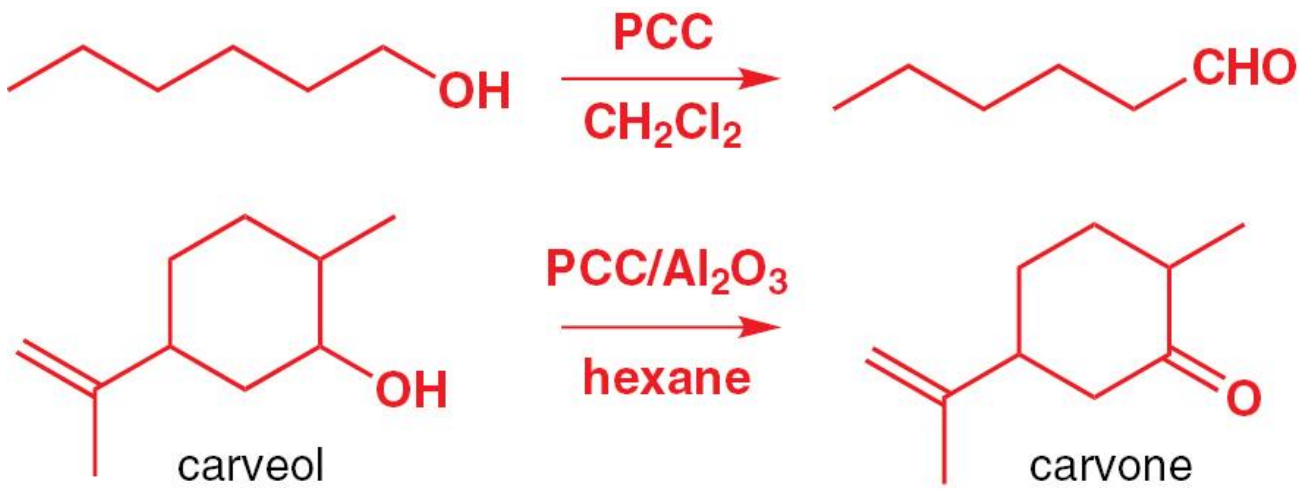
Oxidation of alcohols: Back to aldehydes and ketones

Examples



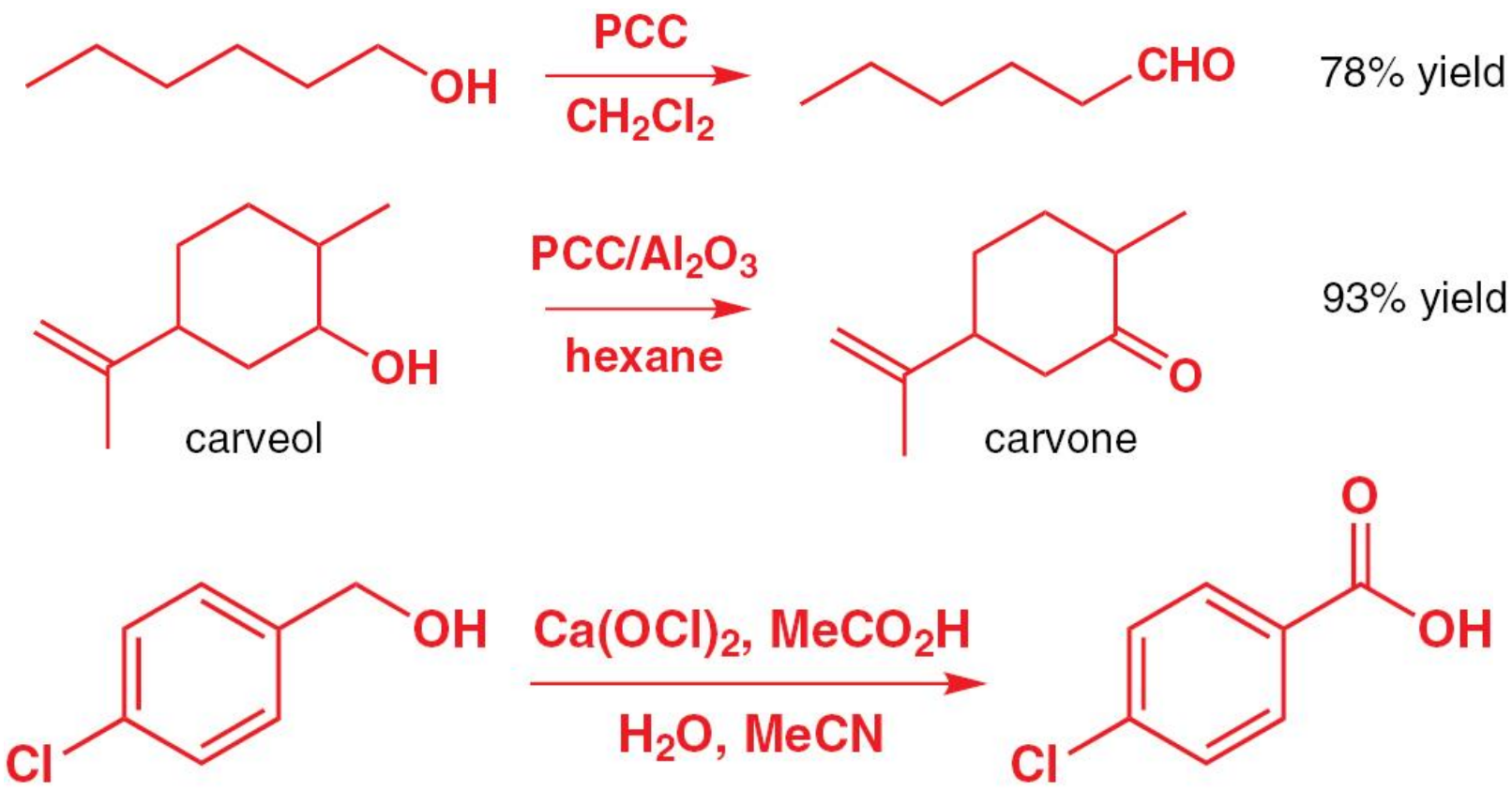
Oxidation of alcohols: Back to aldehydes and ketones

Examples



Oxidation of alcohols: Back to aldehydes and ketones

Examples



How to make organometallics?

- Oxidative insertion of Mg into alkyl halides



How to make organometallics?

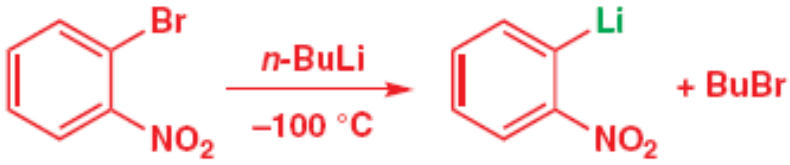
- Oxidative insertion of Mg into alkyl halides



- Oxidative insertion of Li into alkyl halides



- Halogen–metal exchange



How to make organometallics?

- Oxidative insertion of Mg into alkyl halides



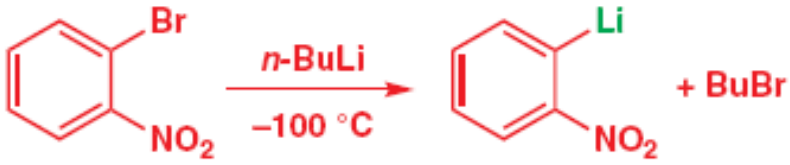
- Oxidative insertion of Li into alkyl halides



- Deprotonation of alkynes



- Halogen–metal exchange



How to make organometallics?

- Oxidative insertion of Mg into alkyl halides



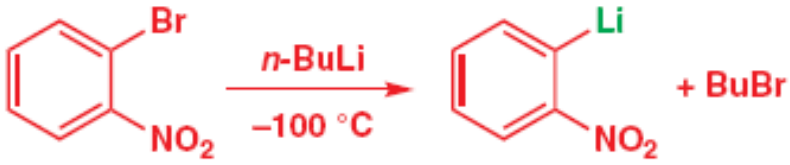
- Oxidative insertion of Li into alkyl halides



- Deprotonation of alkynes



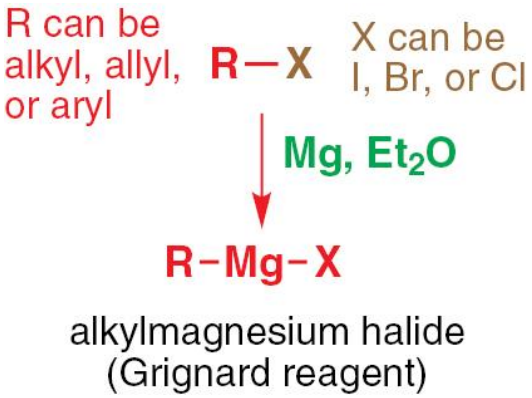
- Halogen–metal exchange



- Transmetallation



How to make Grignard reagents?

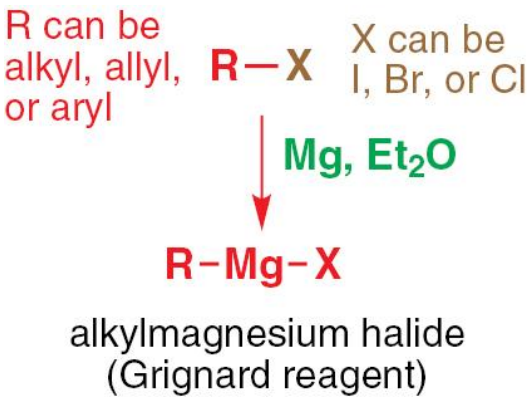


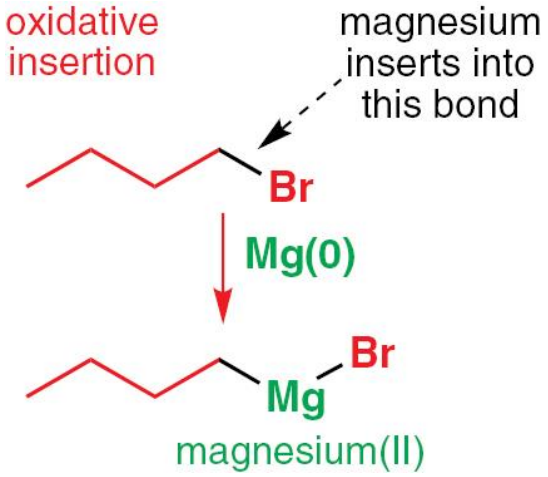
exothermic

or

endothermic

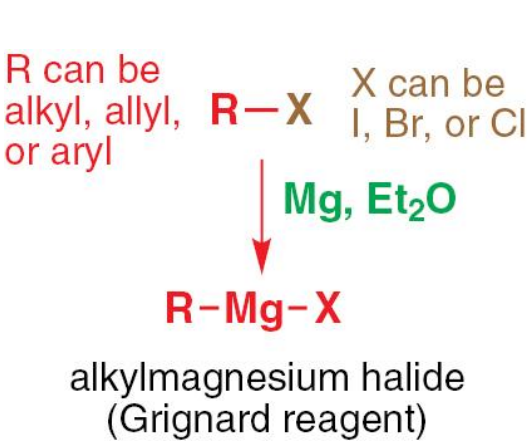
How to make Grignard reagents?

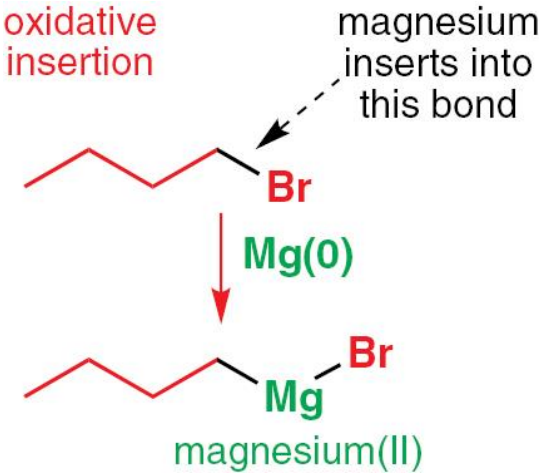




Examples

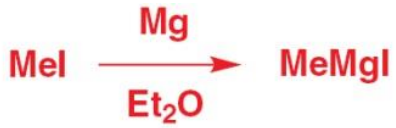
How to make Grignard reagents?

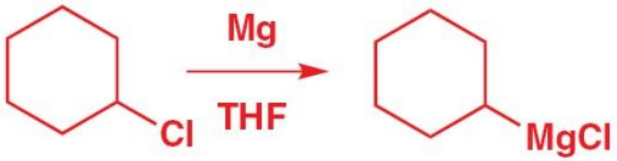


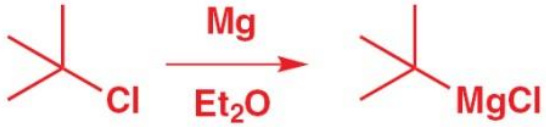


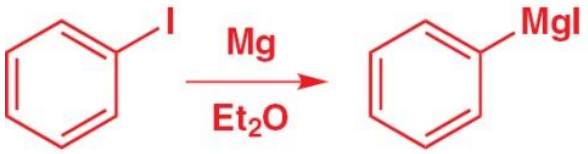
Examples

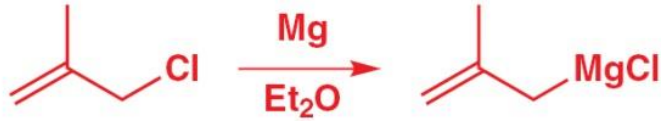






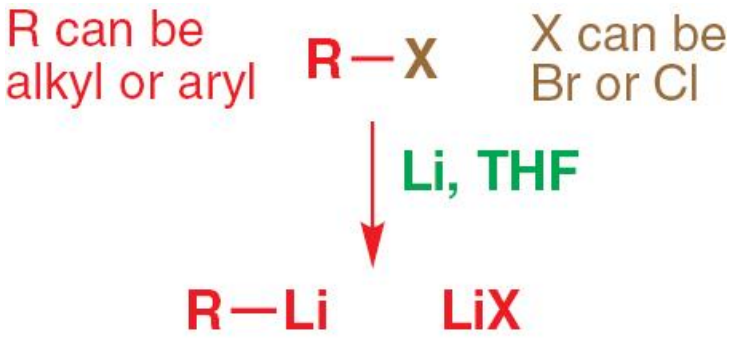






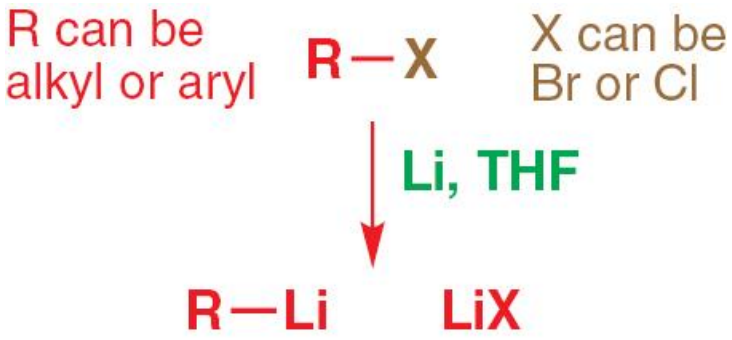


How to make organolithium reagents RLi?



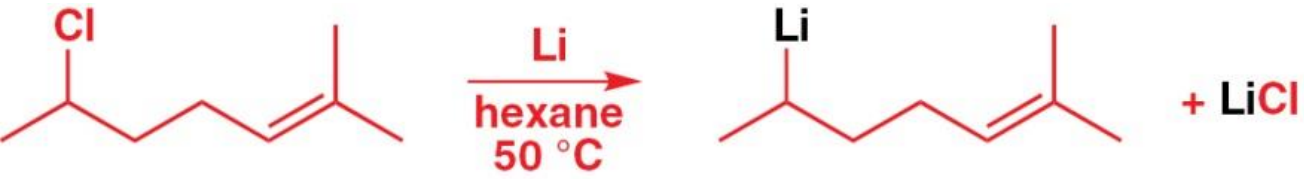
Examples

How to make organolithium reagents RLi?



alkyllithium plus lithium halide

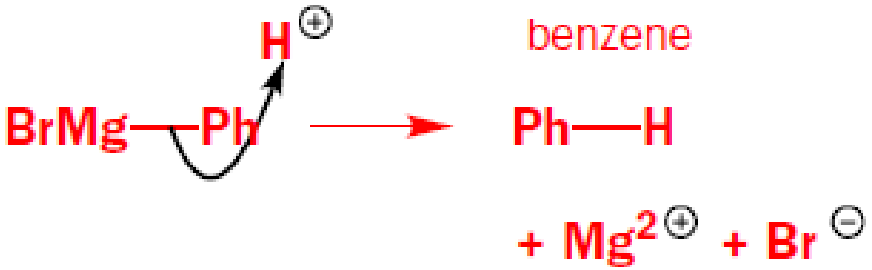
Examples

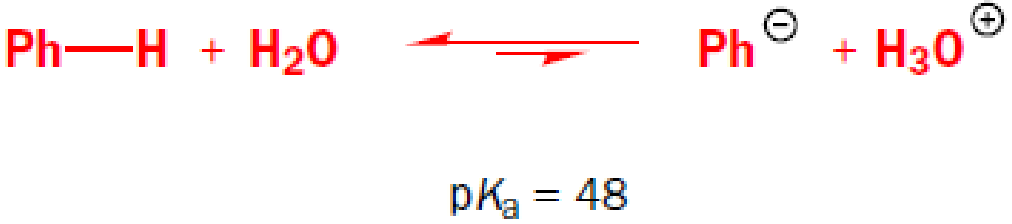




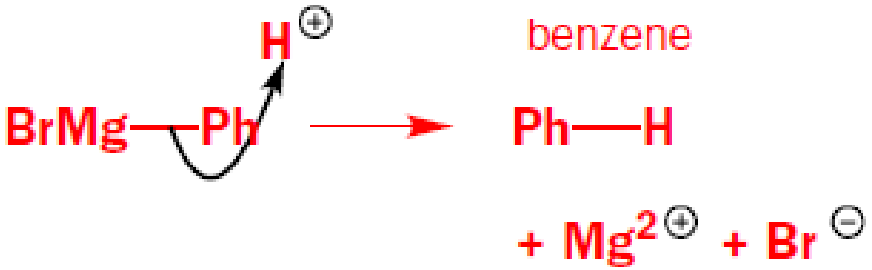


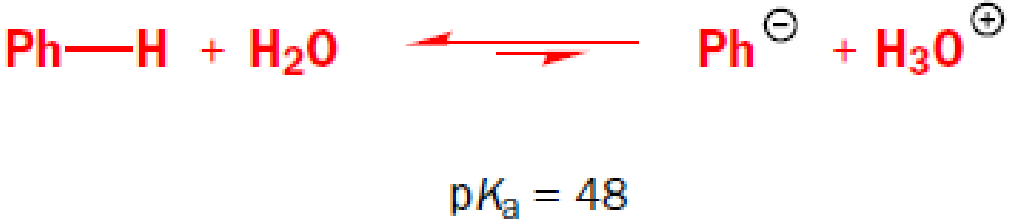
Organometallics as bases

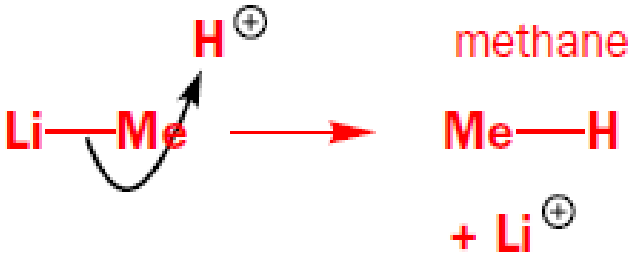


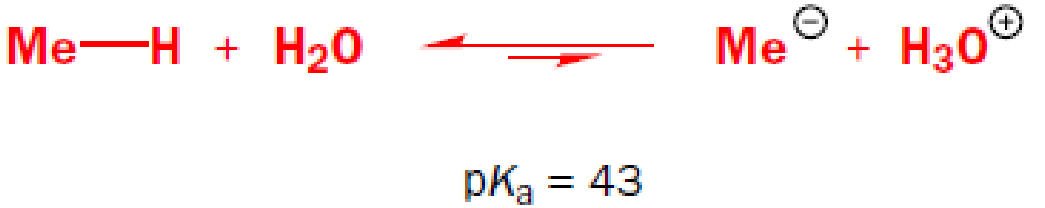


Organometallics as bases



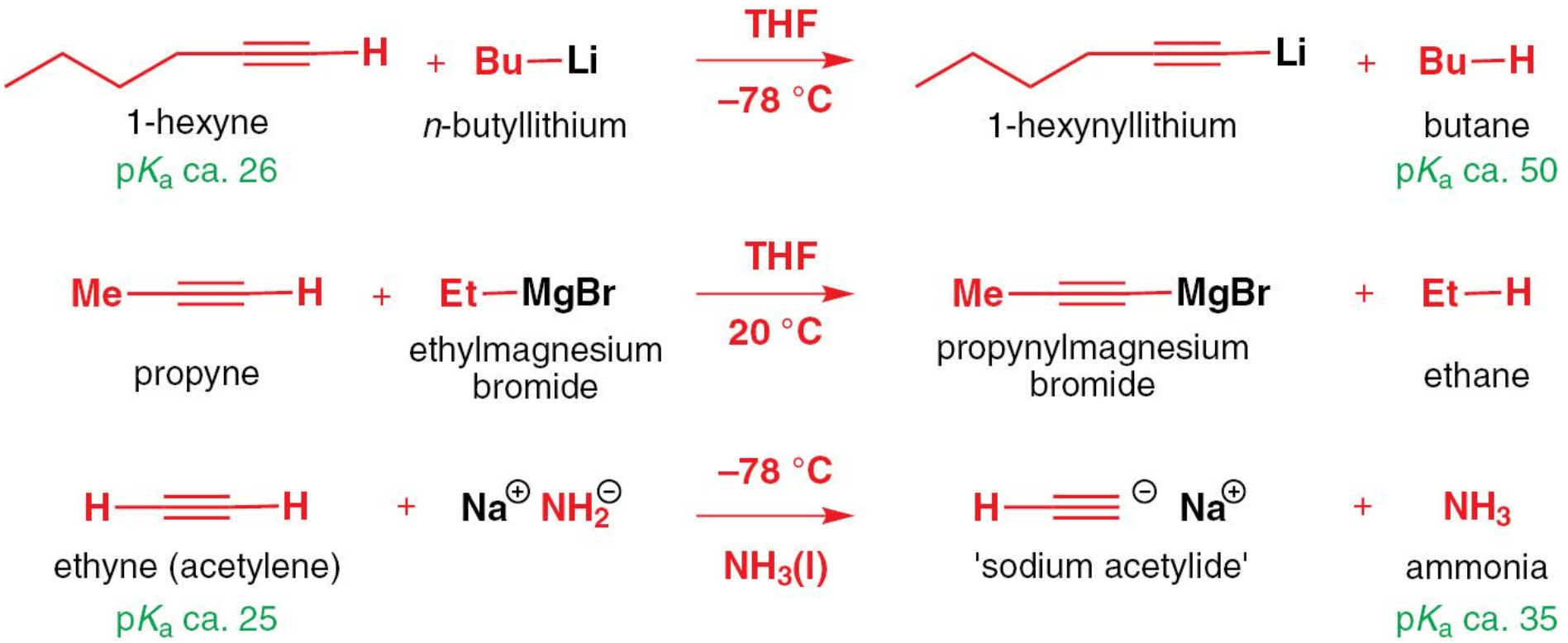






Making Organometallics by deprotonating alkynes

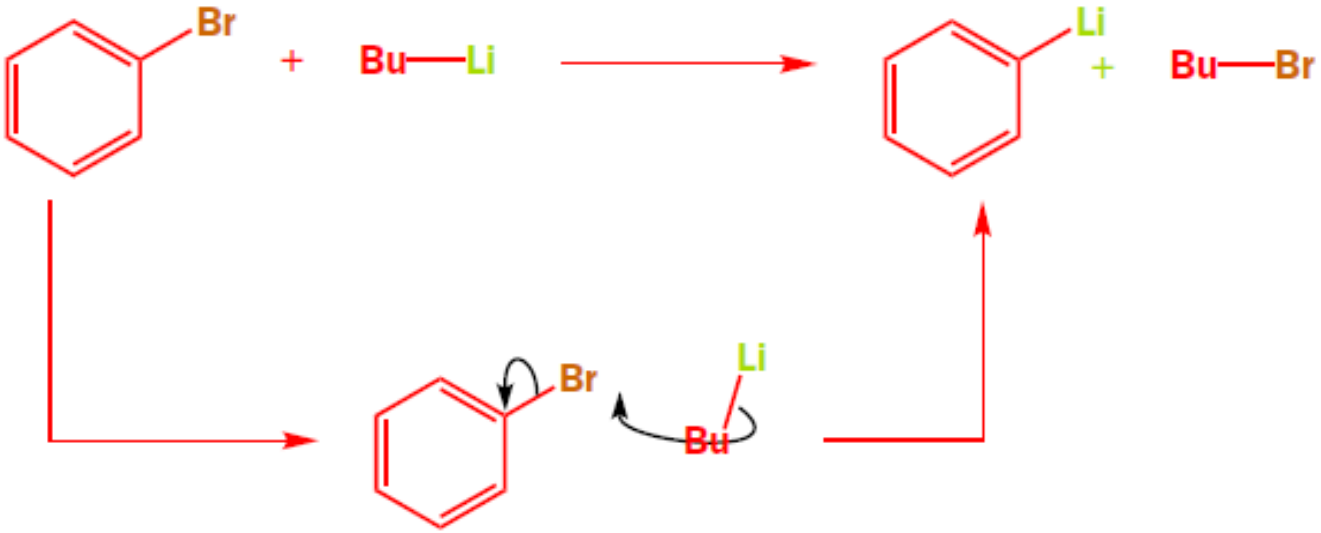
Examples



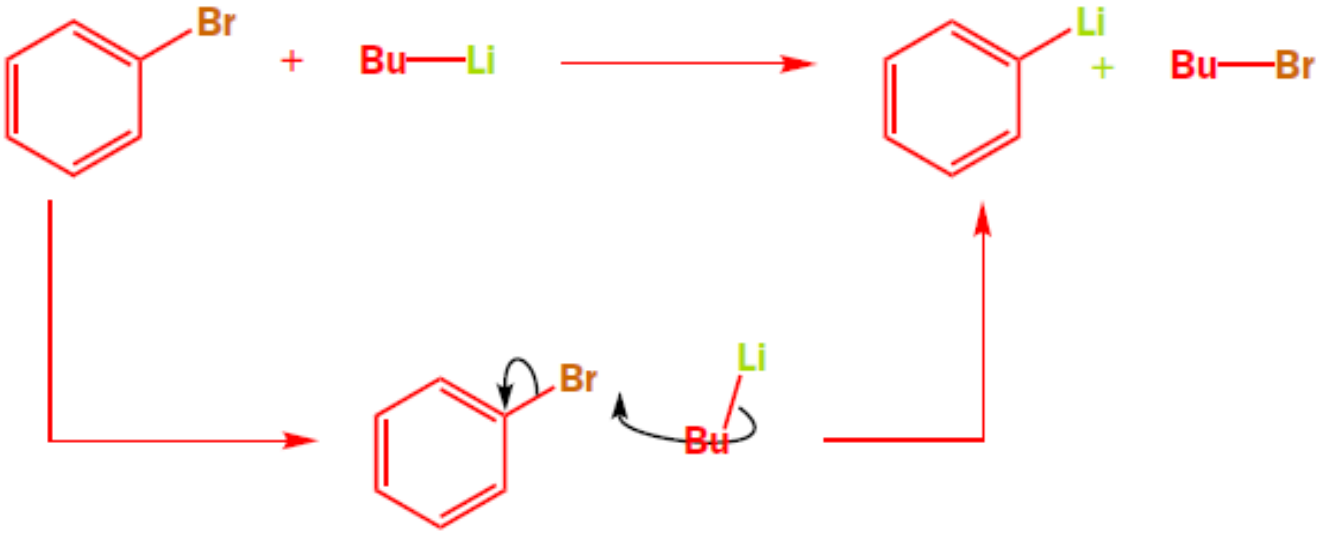
Halogen-metal exchange



Halogen-metal exchange

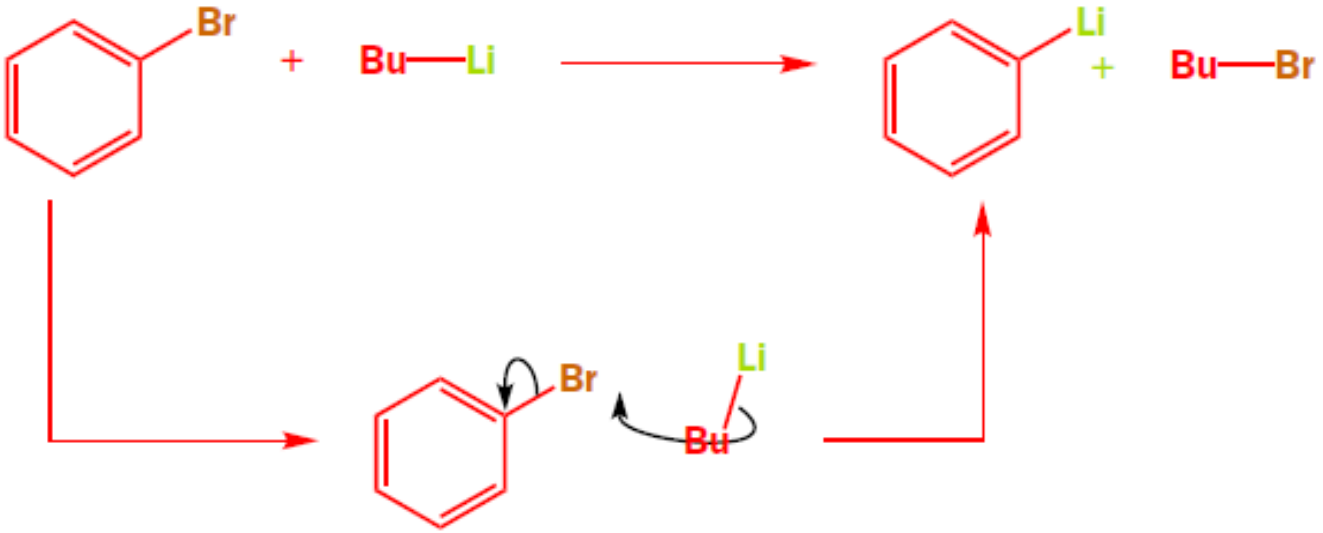


Halogen-metal exchange

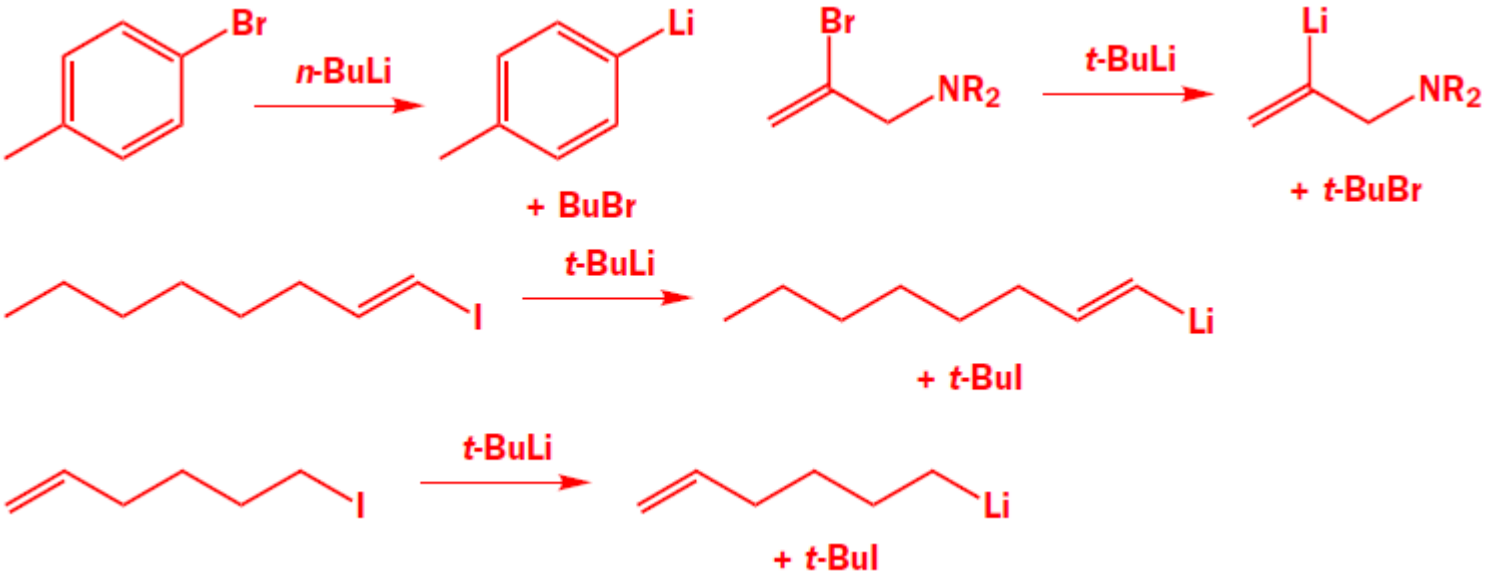


Examples

Halogen-metal exchange



Examples



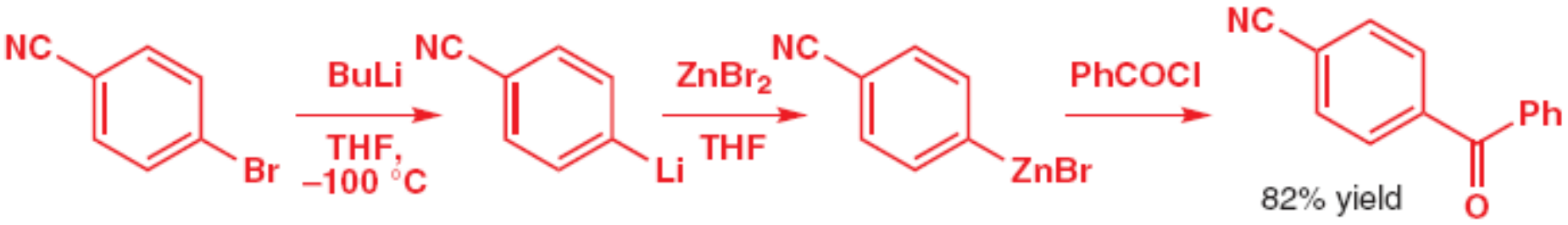
Transmetallation



Transmetallation



Examples



Summary

- **Organometallics: nucleophilic and often strongly basic**
- **Making organometallics from halocompounds**
- **Making organometallics by deprotonating carbon atoms**
- **Using organometallics to make new C-C bonds from C=O groups**